

SOFTMAX REPARAMETERIZATION FOR OUTPUT-HEAD QUANTIZATION

Asim Kadav, Christian Flores, Chirag Arora, Varun Kotte, Hongbo Zheng, Lan Yan,
Priya Shanmugasundaram, Tracy Holloway King

Adobe SDC

{akadav, christanf, charora, vkotte, hongboz, lany, priyash, tking}@adobe.com

ABSTRACT

Large vocabularies make output heads a substantial inference cost in small language models. We propose *softmax reparameterization*, a post-training method that selects a functionally equivalent output head before quantization. The method subtracts a scalar multiple of the vocabulary-row mean from every output row and selects the coefficient by validation KL separately for RTN, activation-weighted MSE, and full-Hessian GPTQ. This one-dimensional search includes the original head and fixed mean-centering, preserves the full-precision softmax distribution, and leaves the trained decoder unchanged; a rank-one correction handles nonlinear logit paths such as soft-capping. Across seven heads, W4 gains concentrate where baseline quantization substantially distorts predictions: on Phi-4-mini, AW-MSE KL falls from 0.936 to 0.256. The gains survive stronger GPTQ calibration and remain complementary to exact per-channel scaling and affine quantization. Across four heads and three W4 quantizers, frozen WikiText-selected coefficients also transfer to C4 and OpenWebMath, outperforming mean-centering in all 18 comparisons where the frozen coefficient differs from 1 and matching it in the remaining six. At W2, used as a compression stress test, benefits broaden across nearly the full model-quantizer matrix. Matched residual analysis shows that improved fidelity can accompany greater logit reconstruction error while reducing the residual’s Fisher-weighted cost. For shift-compatible heads, reparameterization adds no inference operation and preserves packed W4 execution: with the decoder held in BF16, quantizing the Phi output head reduces batch-one generation latency by 10.8% relative to the BF16-head baseline.

1 INTRODUCTION

Modern small language models (SLMs) pair modest decoder sizes with large multilingual vocabularies. For vocabulary size V and hidden width d , the output projection contains Vd weights. In recent 3–4B models, output heads contain roughly 0.4–0.7 billion weights, or about 10–17% of nominal model size (Appendix H.1). Unlike input embeddings, which retrieve selected rows, the dense output projection evaluates the full vocabulary at every decoding step (Grave et al., 2017).

Vocabulary is itself a scaling dimension: scaling-law studies find that useful vocabulary capacity grows with model scale and that larger vocabularies can improve quality (Tao et al., 2024; Huang et al., 2025). Figure 1 shows that modern SLMs already deploy vocabularies far larger than historical 32K-scale designs, and each enlargement scales the head: the output projection performs about Vd multiply-accumulates and reads about $2Vd$ bytes of weights per decoding step at bfloat16 (BF16) precision. Because single-token decoding is memory-bound, this weight movement is a direct cost, and low-bit kernels reduce it in proportion to the stored precision (IST-DASLab, 2025), motivating output-head compression as vocabularies grow.

Despite this cost, practical post-training quantization recipes often retain the output head at higher precision (vLLM Project, 2026). The head directly determines next-token probabilities, without subsequent learned layers to absorb its error. In tied models, its source matrix also serves as the input embedding, introducing a separate deployment constraint (Kurtic et al., 2023). Leaving the

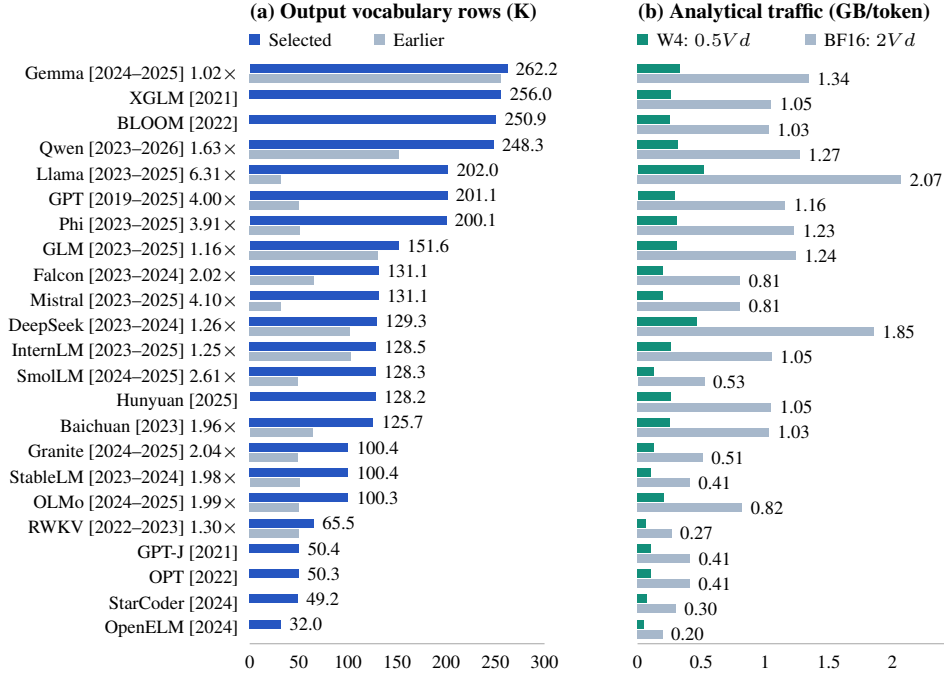


Figure 1: **Large vocabularies create substantial output-head traffic.** (a) Output rows across 23 selected model families, including small models such as Phi, SmolLM, Granite and StableLM. Brackets give earlier–selected release years; ratios compare selected with earlier vocabulary size. Pairs are selected examples, not monotonic trajectories; missing earlier bars indicate no comparison. (b) Analytical weight traffic for the selected checkpoint at batch-one decoding: $2Vd$ bytes in BF16 versus $0.5Vd$ in W4, excluding scales and cache effects. This is a storage-precision comparison, not a measurement of W4 quality or speed. Checkpoint conventions and sources are described in Appendix H.1.

head unchanged therefore preserves a substantial per-token weight read even when the decoder is aggressively compressed.

Output-head quantization is often posed as preserving weights or logits, but softmax depends only on relative logits. Equivalent full-precision output heads can incur different quantization residuals (Figure 2). We exploit this freedom with *softmax reparameterization*, selecting an equivalent representative before quantization.

Function-preserving reparameterizations are widely used in quantization, including channel scaling and rotations (Xiao et al., 2023; Ashkboos et al., 2024; Liu et al., 2025), while SageAttention applies fixed mean-centering to attention keys before quantization (Zhang et al., 2024). Output Embedding Centering uses vocabulary-row mean-centering during pretraining (Stollenwerk et al., 2026). Our contribution is a post-training scalar search within the output head’s additive softmax equivalence class, selecting the representative by post-quantization predictive fidelity.

Let μ be the vocabulary-row mean of the output head. We search the scalar family

$$W_t = W - t\mathbf{1}\mu^\top$$

and select t by the KL divergence of the quantized head to the source model on validation data. The search uses pretrained weights and requires no retraining or changes to the decoder. The search includes the original head ($t = 0$) and ordinary mean-centering ($t = 1$), and is performed separately for each base quantizer.

The largest gains occur in heads with high baseline quantization error. At W4, several heads are already close to the source distribution and therefore have little error to recover, while Phi-4-mini and the large-vocabulary BLOOM and XGLM heads exhibit large gains. Under more aggressive W2 quantization this distinction largely disappears: every evaluated head is substantially perturbed, and

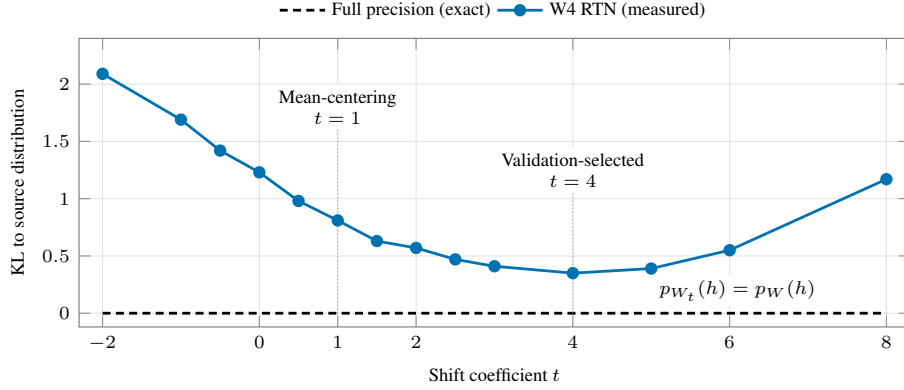


Figure 2: **Quantization fidelity across equivalent output-head parameterizations.** Shared row shifts preserve predictions exactly (dashed: algebraic KL = 0), but change Phi-4-mini W4 RTN test KL (blue). Mean-centering uses $t = 1$; $t = 4$ is selected on separate validation articles. All shifts use the same quantizer and test states.

reparameterization improves nearly every model and quantizer combination across RTN, AW-MSE and GPTQ (Table 1). Representative selection improves fidelity across multiple model families, with gains extending to more heads under W2 quantization.

In our packed Phi deployment, W4 head compression reduces end-to-end batch-one latency by 10.8% and batch-16 latency by 9.4%, and reparameterization recovers quality without sacrificing this speedup. The improvement itself does not come from better weight reconstruction: the selected representative can carry more logit error while placing it where the softmax barely responds, an effect examined in Section 4.

Our contributions are threefold:

- **Softmax reparameterization.** We introduce a post-training search over exactly equivalent output-head parameterizations, selecting the representative that best preserves prediction fidelity under the deployed quantizer. This differs from fixed mean-centering and concurrent symmetry-based quantization methods by searching a one-dimensional path within the additive softmax equivalence class of the frozen output head. A rank-one correction extends the construction to nonlinear logit paths.
- **Low-bit recovery across models and quantizers.** At W4, reparameterization substantially improves heads with large baseline quantization error across Phi, BLOOM/BLOOMZ and XGLM, while already faithful heads change little. At W2, where all evaluated heads become substantially distorted, the benefit broadens across essentially the full model and quantizer matrix. The effect persists under RTN, AW-MSE and full-Hessian GPTQ and reproduces on untouched Phi and BLOOM holdouts.
- **Deployment and mechanism.** Packed W4 head compression reduces batch-one generation latency by 10.8%, and reparameterization recovers fidelity without sacrificing this speedup. Residual analysis explains why: the selected representative can increase ordinary reconstruction error while shifting that error into directions that matter less to the predictive distribution.

2 SOFTMAX REPARAMETERIZATION

2.1 EXACT OUTPUT-HEAD EQUIVALENCE

The linear-softmax output head admits an exact additive reparameterization. Softmax depends only on relative logits, so subtracting the same vector from every vocabulary row leaves the output distribution unchanged. This gives a family of equivalent heads whose quantized fidelity can differ.

Let V be the vocabulary size and d the hidden width. The output head $W \in \mathbb{R}^{V \times d}$ maps a final hidden state $h \in \mathbb{R}^d$ to logits $z = Wh$, giving next-token probabilities $p_W(h) = \text{softmax}(Wh)$. Let

$\mathbf{1} \in \mathbb{R}^V$ denote the all-ones vector. For any shared shift $a \in \mathbb{R}^d$, define $W_a = W - \mathbf{1}a^\top$. Then

$$W_a h = Wh - (a^\top h)\mathbf{1}, \quad p_{W_a}(h) = p_W(h). \quad (1)$$

The equality follows from cancellation of a common exponential factor in softmax. It is an exact symmetry of a linear-softmax readout, independent of the hidden-state distribution. An unchanged output bias also preserves this identity. The shared row component is unidentifiable from prediction probabilities.

Let $\mathcal{B}$ denote a fixed base quantization procedure, including its precision, grouping, range objective and solver, and let $\mathcal{B}(W_a)$ denote its reconstructed weight matrix. In general, $\mathcal{B}(W_a)$ and $\mathcal{B}(W)$ are not related by a vocabulary-wide shift. Define the weight-quantization residual $E_a = \mathcal{B}(W_a) - W_a$. Equation 1 gives

$$p_{\mathcal{B}(W_a)}(h) = \text{softmax}(Wh + E_a h). \quad (2)$$

The induced logit error is $E_a h$. Reparameterization changes the quantization residual while preserving the source distribution, but equivalence alone does not guarantee improved fidelity.

2.2 FIDELITY-SELECTED REPARAMETERIZATION

Low-bit quantization of the output head can substantially distort next-token probabilities. To preserve output fidelity, we search over shared shifts of the head weights before quantization and select the shift by validation KL. Specifically, we restrict the d -dimensional shift to the vocabulary-row mean direction:

$$\mu = \frac{1}{V} W^\top \mathbf{1}, \quad W_t = W - t\mathbf{1}\mu^\top, \quad Q_t = \mathcal{B}(W_t; \mathcal{H}_{\text{fit}}). \quad (3)$$

The vocabulary-row mean identifies the shared component removed by ordinary mean-centering. Varying its magnitude gives a low-cost, empirically effective one-dimensional search within the exact equivalence class; we do not claim that this path contains the optimal representative. Figure 2 illustrates this search on Phi-4-mini: full-precision predictions remain unchanged across shifts, while quantized KL varies substantially.

The fitting states $\mathcal{H}_{\text{fit}}$ are used only to fit the base quantizer, while a disjoint set $\mathcal{H}_{\text{val}}$ selects the shift. Quantizer-specific fitting details are given in Appendix A.

For a finite grid $\mathcal{T}$ containing zero (the 14-point grid is specified in Appendix A.1), the proposed selection rule is

$$t^* = \arg \min_{t \in \mathcal{T}} L_{\text{val}}^{\mathcal{B}}(t), \quad L_{\text{val}}^{\mathcal{B}}(t) = \frac{1}{|\mathcal{H}_{\text{val}}|} \sum_{h \in \mathcal{H}_{\text{val}}} D_{\text{KL}}(p_W(h) \| p_{Q_t}(h)). \quad (4)$$

We select by KL to preserve the source model’s next-token distribution. Perplexity can improve through a change in confidence even when predictions depart further from the source (Appendix I). Final evaluation uses articles disjoint from quantizer fitting and shift selection.

Because $0 \in \mathcal{T}$, exact evaluation of Equation 4 gives $L_{\text{val}}^{\mathcal{B}}(t^*) \leq L_{\text{val}}^{\mathcal{B}}(0)$. This guarantee applies only to validation KL under the quantizer used for selection; it does not extend to unseen data or a different quantizer.

2.3 WHY FIXED MEAN-CENTERING IS INSUFFICIENT

A natural choice is to subtract the vocabulary-row mean, corresponding to $t = 1$. This minimizes the full-precision weight norm, but does not necessarily preserve predictions best after quantization. We therefore treat mean-centering as a candidate in the search and allow validation KL to select a different shift.

Writing $W_c = W - \mathbf{1}\mu^\top$, the sum of its row vectors is zero ($\mathbf{1}^\top W_c = \mathbf{0}^\top$), so

$$\|W_t\|_F^2 = \|W_c\|_F^2 + V(1-t)^2 \|\mu\|_2^2. \quad (5)$$

Changing t also changes clipping ranges and rounding. On Phi, validation KL selects $t = 4$. Appendix E.2 reports a separate scan in which raw and projected weight MSE both select $t = 1$, while diagnostic KL favors other coefficients.

Table 1: **Output-head quantization fidelity relative to the BF16 source model.** Seven heads at W4 and W2 under RTN, AW-MSE and full-Hessian GPTQ. Each cell reports test KL before ($t = 0$) $\rightarrow$ after reparameterization (t^*); lower is better. The shift is selected separately for each quantizer by validation KL. Gemma 4 ($\dagger$) uses the rank-one soft-cap correction.

Model	W4 KL			W2 KL		
	RTN	AW-MSE	GPTQ	RTN	AW-MSE	GPTQ
Phi-4-mini	1.23 $\rightarrow$ 0.351	0.936 $\rightarrow$ 0.256	0.158 $\rightarrow$ 0.059	98.2 $\rightarrow$ 74.2	41.6 $\rightarrow$ 16.2	3.08 $\rightarrow$ 1.16
Gemma 3	0.049 $\rightarrow$ 0.046	0.041 $\rightarrow$ 0.038	0.033 $\rightarrow$ 0.033	2.66 $\rightarrow$ 2.21	0.822 $\rightarrow$ 0.714	0.508 $\rightarrow$ 0.491
Gemma 4 $\dagger$	0.012 $\rightarrow$ 0.010	0.008 $\rightarrow$ 0.008	0.008 $\rightarrow$ 0.008	0.911 $\rightarrow$ 0.911	0.212 $\rightarrow$ 0.199	0.165 $\rightarrow$ 0.162
Qwen3.5	0.021 $\rightarrow$ 0.019	0.013 $\rightarrow$ 0.013	0.011 $\rightarrow$ 0.010	1.84 $\rightarrow$ 1.74	0.312 $\rightarrow$ 0.285	0.200 $\rightarrow$ 0.197
BLOOM-1.7B	1.12 $\rightarrow$ 0.531	0.594 $\rightarrow$ 0.136	0.037 $\rightarrow$ 0.028	16.7 $\rightarrow$ 6.15	4.46 $\rightarrow$ 1.46	0.427 $\rightarrow$ 0.335
BLOOMZ-1.7B	1.12 $\rightarrow$ 0.483	0.662 $\rightarrow$ 0.158	0.042 $\rightarrow$ 0.032	16.2 $\rightarrow$ 6.22	3.99 $\rightarrow$ 2.12	0.456 $\rightarrow$ 0.367
XGLM-1.7B	2.13 $\rightarrow$ 0.143	0.584 $\rightarrow$ 0.095	0.009 $\rightarrow$ 0.007	97.8 $\rightarrow$ 55.3	23.7 $\rightarrow$ 3.73	0.173 $\rightarrow$ 0.128

Applicability and inference. For a shift-compatible logit path, the selected shift is folded into the packed head with no additional inference operation. Elementwise nonlinearities such as tanh soft-capping require restoring $t(\mu^\top h)\mathbf{1}$ before the nonlinearity; Appendix B gives the construction and equivalence check. Tied models retain the source input embedding and quantize a separate output copy (Kurtic et al., 2023). Appendix G.2 details the storage implications; Algorithm 1 in the appendix specifies the search procedure.

3 EXPERIMENTS

Experimental setup. We quantize only the output head while keeping the decoder fixed. Each comparison measures how closely the quantized head reproduces its own source model’s predictions.

We evaluate seven heads: modern SLMs (Gemma 3/4, Qwen3.5 and Phi) and three additional large-vocabulary models (BLOOM-1.7B, BLOOMZ-1.7B and XGLM-1.7B). The W4 comparison uses G128 groups, BF16 decoders, and RTN, AW-MSE or full-Hessian GPTQ (Table 1). Every head first passes the applicability check of Appendix B: the linear-softmax heads reproduce the source to $\text{KL} \lesssim 10^{-9}$, while Gemma 4 uses the rank-one soft-cap correction of Equation 9 (shifted BF16 KL $\approx 7 \times 10^{-10}$, source PPL 66.41).

Fitting uses 128 WikiText articles with eight states each; validation and test use 16 disjoint articles each. For each base quantizer, we select t^* from a 14-point grid over $[-2, 8]$ by validation KL, then report source-to-candidate KL and PPL from FP32 readouts on BF16-decoder states. The selection and test articles were encountered during earlier exploration; a separate untouched holdout is reported below.

KL values are token averages for fixed quantized heads. Key ablations include article- or document-bootstrap confidence intervals; the BLOOM stability study uses ten validation-subsampling seeds. These quantify evaluation-sample uncertainty and coefficient-selection stability, respectively; they do not measure sensitivity to alternative calibration samples or refitted quantizers.

3.1 OUTPUT-HEAD QUANTIZATION RESULTS

When W4 quantization substantially perturbs the output distribution, reparameterization can recover most of the lost fidelity. On XGLM, RTN test KL falls from 2.13 to 0.143 (93%). Under AW-MSE, KL falls from 0.936 to 0.256 on Phi (73%), from 0.594 to 0.136 on BLOOM (77%), and from 0.662 to 0.158 on BLOOMZ (76%). Full-Hessian GPTQ has substantially lower baseline KL, but reparameterization still improves these heads (Table 1). Gemma 3/4 and Qwen3.5 already have low W4 baseline KL and change little.

At W2, baseline distortion rises across every evaluated head, and the benefit broadens across nearly the full model–quantizer matrix. We use W2 as a compression stress test; absolute quality remains poor for several configurations despite the KL reductions. Appendix C.4 reports the full W3 and W2 results.

Table 2: **Fixed mean-centering versus validation-selected reparameterization at W4.** Test KL relative to the BF16 source model for the unshifted head ($t = 0$), mean-centered head ($t = 1$), and validation-selected head (t^*), using the Table 1 evaluation split. Lower is better.

Head	RTN			AW-MSE			GPTQ		
	$t=0$	$t=1$	t^*	$t=0$	$t=1$	t^*	$t=0$	$t=1$	t^*
Phi-4-mini	1.229	0.811	0.351	0.936	0.577	0.256	0.158	0.117	0.059
BLOOM-1.7B	1.117	0.978	0.531	0.594	0.451	0.136	0.037	0.036	0.028
BLOOMZ-1.7B	1.119	0.859	0.483	0.662	0.399	0.158	0.042	0.039	0.032
XGLM-1.7B	2.129	0.143	0.143	0.584	0.095	0.095	0.009	0.007	0.007

3.2 ABLATION STUDIES AND VALIDATION

Comparison with fixed mean-centering. The test curve in Figure 2 is a fixed-centering control. On Phi, $t = 1$ lowers RTN test KL from about 1.23 to 0.81, but the validation-selected $t = 4$ lowers it further to about 0.35, roughly 57% below fixed mean-centering. KL rises again at $t = 5, 6, 8$, and all sampled negative shifts worsen the head, giving an interior minimum on the sampled grid. Mean-centering ($t = 1$) minimizes the full-precision weight norm, but need not minimize KL after quantization. On Phi, the validation-selected $t = 4$ gives lower test KL despite a larger weight norm.

Table 2 shows that validation-selected shifts outperform fixed mean-centering on Phi, BLOOM and BLOOMZ. On XGLM, all three quantizers select mean-centering ($t^* = 1$).

Because the preferred representative also depends on the base quantizer, we select t separately for each quantizer throughout; transferring an RTN-selected coefficient to AW-MSE can reduce fidelity (Appendix E.1).

Calibration, scaling, and affine quantization. Reparameterization also improves full-Hessian GPTQ: on Phi, validation-selected shifting reduces W4 test KL by 62%, with smaller reductions on BLOOM, BLOOMZ and XGLM (Table 1). We then test whether shifting still improves fidelity with more calibration data, exact channel scaling, or integer zero points.

On Phi, increasing GPTQ calibration from 1,024 to 65,536 states, with the range initializer held fixed, lowers raw W4 test KL from 0.160 to 0.089, yet reparameterization further reduces it to 0.035 (61%). Exact per-channel scaling also substantially improves the unshifted head, but the additive shift remains complementary: KL falls from 0.240 to 0.172 under scaled AW-MSE and from 0.337 to 0.242 under scaled RTN. Likewise, affine RTN with integer zero points improves from 0.750 to 0.259 after reparameterization. Paired article-bootstrap 95% intervals exclude zero for each of these gains. Appendix D reports the full calibration, scaling, affine and direction controls.

Evaluation on an independent holdout. The preceding evaluation articles were disjoint from fitting and validation but had been encountered during earlier exploration. We therefore froze the grid, selection rule, quantizers and code and evaluated once on 26 previously unused WikiText articles. On Phi, the selected representative reduces AW-MSE KL by 73.6% (0.943 to 0.249) and RTN KL by 71.1%; article-bootstrap 95% intervals are [72.9, 74.4]% and [70.4, 71.7]%, respectively. We repeat the same protocol on BLOOM-1.7B: AW-MSE KL falls by 74.5% (0.558 to 0.142; 95% interval [72.4, 76.2]%) and RTN by 51.7%. The large reductions reproduce on untouched data for both Phi and BLOOM. Full likelihood and audit statistics appear in Appendix C.1.

Cross-domain transfer and coefficient stability. We freeze the WikiText-selected coefficients and evaluate them without retuning on C4 and OpenWebMath. Across four heads, three W4 quantizers and two domains, the frozen coefficient outperforms fixed mean-centering in all 18 comparisons where the frozen WikiText-selected coefficient differs from 1, and ties the remaining six XGLM cases where it equals 1. In a separate BLOOM study, increasing validation size to 1,024 documents yields identical selected coefficients across ten seeds for each C4/OpenWebMath-quantizer pair; all selected coefficients remain different from 1. The preferred coefficient is nevertheless distribution-dependent: larger C4 and OpenWebMath validation sets can select different coefficients from WikiText (Appendix C.3).

Table 3: **Prediction fidelity and generation latency with packed W4 output heads on Phi-4-mini.** Quality is evaluated on identical WikiText tokens. KL is measured relative to BF16 using FP32 readout; perplexity uses vLLM-packed execution. A10G latency covers 64-token greedy generation at batch sizes 1 and 16, with prefix caching disabled. Dashes indicate unmeasured GPTQ latency. Timing protocol: Appendix G.

Head	KL to BF16	WikiText PPL	B1 (ms)	B16 (ms)
BF16	0.00	11.65	1136.6	1335.8
W4 min-max	1.23	40.57	1015.3	1211.8
W4 min-max + shift	0.34	16.28	1014.1	1210.1
W4 AW-MSE	0.97	31.65	1014.8	1210.2
W4 AW-MSE + shift	0.28	15.40	1013.4	1210.3
W4 GPTQ	0.15	13.67	—	—
W4 GPTQ + shift	0.06	12.23	—	—

Appendix J explores a higher-capacity group-wise extension, with additional gains under W4 RTN and AW-MSE in a nine-model study and a separate SmolLM3 evaluation with matched search budgets.

3.3 PACKED INFERENCE EVALUATION

The preceding experiments evaluate quantized heads on captured hidden states. We next test whether the fidelity gains persist in deployed generation and measure the latency benefit of head compression. Here, packed W4 execution stores weights as 4-bit codes and computes the output projection using the Marlin INT4 kernel (IST-DASLab, 2025). We export zero-shift and $t = 4$ Phi heads from the same BF16 source and evaluate them through vLLM (Kwon et al., 2023), preserving all non-head tensors.

On 16,352 matched WikiText tokens, the shift lowers AW-MSE perplexity from 31.65 to 15.40 and KL to BF16 from 0.97 to 0.28 (Table 3). It also improves our strongest deployed baseline, full-Hessian GPTQ (Frantar et al., 2023): perplexity falls from 13.67 to 12.23 and KL from 0.15 to 0.06, leaving approximately 5% perplexity excess over BF16’s 11.65. Thus representative selection adds fidelity even after GPTQ has compensated rounding error, with the resulting head using the same INT4 format.

The shift is folded into the weights before quantization, preserving the packed layout and adding no inference operation for Phi’s linear-softmax head. Measured min-max and AW-MSE heads have essentially unchanged latency after shifting; shifted AW-MSE has 10.8% lower batch-one latency than BF16. GPTQ uses the same Marlin format; its latency was not measured.

The tied Phi deployment retains its BF16 input embedding and adds a separate packed output head, increasing resident model weights from 7.17 to 7.47 GiB. This memory cost applies equally to shifted and unshifted W4 heads. Appendix G.1 gives the numerical and timing protocols; Appendix G.2 details the storage trade-off.

4 ANALYSIS OF QUANTIZATION ERROR

We analyze how representative selection changes the quantization residual and its effect on prediction fidelity. Reconstruction-based quantizers such as GPTQ measure weight error through its effect on layer outputs (Frantar et al., 2023), whereas KL also depends on the direction of the logit perturbation relative to the source distribution. SoftWater similarly incorporates feature covariance and softmax curvature into output-head quantization (Cavalcanti & Wilson, 2026).

4.1 SOFTMAX SENSITIVITY TO LOGIT ERROR

We use the softmax Fisher matrix to quantify how logit errors affect the predicted distribution. For small perturbations, its quadratic form approximates KL divergence, accounting for both the

magnitude and direction of the error. This lets us examine why a shift can reduce KL even when total logit-error energy increases.

For a fixed shift t , let $E = E_t = \mathcal{B}(W_t) - W_t$ be the weight residual and $e = Eh$ the induced logit error. Write $p = p_W(h)$ for the source distribution and $q = \text{softmax}(Wh + e)$ for the perturbed distribution. The forward KL, $D_{\text{KL}}(p||q) = \sum_{v=1}^V p_v \log(p_v/q_v)$, weights each token’s log-probability ratio by its source probability and is zero when the distributions agree. For small e , it has the local expansion

$$D_{\text{KL}}(p || \text{softmax}(Wh + e)) = \frac{1}{2} e^\top F(p) e + O(\|e\|_2^3), \quad F(p) = \text{diag}(p) - pp^\top, \quad (6)$$

Here $F(p)$ is the Hessian of KL with respect to the logit perturbation at $e = 0$, and $\text{diag}(p)$ places the source probabilities on the diagonal.

Its quadratic form is the source-probability-weighted variance of the logit errors:

$$e^\top F(p) e = \sum_{v=1}^V p_v (e_v - \bar{e}_p)^2, \quad \bar{e}_p = \sum_{v=1}^V p_v e_v.$$

Thus the approximation measures changes in relative logits, weighted by their source probabilities, rather than total squared logit error. The $O(\|e\|_2^3)$ remainder contains higher-order terms; the quadratic is a local approximation whose accuracy we check against measured KL in Section 4.2.

This analysis applies directly to shift-compatible logit paths; for nonlinear pre-softmax transformations such as Gemma 4’s soft-cap, the Fisher quadratic must use the post-transformation logit residual after the rank-one correction (Appendix B). The categorical Fisher matrix satisfies $F(p)\mathbf{1} = 0$, expressing the common-shift invariance of softmax (Martens, 2020). To separate this invisible component, let $P = I - \mathbf{1}\mathbf{1}^\top/V$ project out the vocabulary-wide mean. All equivalent heads satisfy $PW_t = PW$, whereas their quantized representatives need not satisfy $PQ_t = PQ_0$.

The hidden-state second moment $\Sigma = \mathbb{E}[hh^\top]$ determines logit-error energy through $\mathbb{E}\|Eh\|_2^2 = \text{tr}(E\Sigma E^\top)$, while $F(p)$ determines its local distributional cost. AW-MSE uses diagonal moments of Σ when fitting the quantizer. Here, the quantizer’s objective remains fixed, and the Fisher quadratic diagnoses the residuals produced by different representatives; selection uses actual validation KL.

4.2 EMPIRICAL ANALYSIS OF QUANTIZATION RESIDUALS

On Phi-4-mini, mean-centering ($t = 1$) minimizes raw logit-error energy among the three measured coefficients, but the validation-selected $t = 4$ achieves lower KL. Table 4 compares $t \in \{0, 1, 4\}$ on 8,176 held-out states, computing actual KL, Fisher error and probability-bin diagnostics from the same quantized artifact at each coefficient. Numerical conventions are detailed in Appendix F.1.

From $t = 0$ to $t = 4$, raw logit-error energy increases by factors of 3.0 under RTN and 2.7 under AW-MSE, while Fisher-weighted error falls by 72–73%.

The same mechanism persists under stronger full-Hessian GPTQ. Using 65,536 calibration states, the selected $t = 4$ representative increases source-normalized logit-error energy D_z from 0.025% to 0.067% ($2.66\times$), while Fisher-weighted cost (Fisher/2) falls from 0.0888 to 0.0347 and test KL from 0.0894 to 0.0349. The Fisher quadratic matches actual KL within 0.7% at both coefficients, showing that the residual becomes less costly to the predictive distribution even after GPTQ’s second-order error compensation (Appendix F.2).

For RTN and AW-MSE in Table 4, the quadratic at $t = 4$ is within 1.4% and 6.2% of actual KL, respectively. Across all six rows, the relative discrepancy reaches 12.4%. The quadratic tracks the observed improvement despite these finite-perturbation discrepancies.

Fixed source-probability bins reveal where the residual changes. Under AW-MSE, the share of vocabulary-centered error energy on entries with $p < 10^{-6}$ rises from 92.4% to 98.2%; these entries carry only 0.20% of the probability mass. Meanwhile, absolute error energy on entries with $p \geq 0.01$, carrying 84.8% of the mass, falls by more than half. The selected shift thus reduces error on likely outputs even as total reconstruction error increases.

Table 4: **Lower KL despite greater logit error on Phi.** Moving from $t = 0$ to $t = 4$ increases D_z while reducing Fisher/2 and KL. D_z is source-normalized logit-error energy; Common is the fraction removed by vocabulary centering. Fisher/2 denotes $\frac{1}{2}\mathbb{E}[e^\top F(p)e]$. Metrics use matched heads and states.

Quantizer	t	$\ E_t\ /\ W_t\ $	D_z (%)	Common (%)	Fisher/2	Actual KL
RTN	0	0.141	0.239	0.03	1.271	1.223
RTN	1	0.137	0.189	0.00	0.720	0.821
RTN	4	0.150	0.707	5.67	0.347	0.352
AW-MSE	0	0.125	0.249	4.06	0.861	0.935
AW-MSE	1	0.121	0.191	0.13	0.532	0.583
AW-MSE	4	0.136	0.677	27.51	0.240	0.256

The RTN bin measurements and common-component decomposition are reported in Appendix F.1. Across all three quantizers, the selected representative lowers the distributional cost of the residual. Under RTN and AW-MSE, this additionally coincides with a larger softmax-invariant component. These measurements explain the KL reduction on Phi but do not establish which other heads will benefit.

5 RELATED WORK

Quantization and equivalent parameterizations. SmoothQuant redistributes activation and weight scales, AWQ uses activation statistics to choose channel scales, and QuaRot and SpinQuant use rotations to improve low-precision inference (Xiao et al., 2023; Lin et al., 2024; Ashkboos et al., 2024; Liu et al., 2025). These methods exploit function-preserving changes in parameterization to improve quantization. Concurrent GaugeQuant learns quantization-friendly bases from internal Transformer symmetries during training (Bento & Seabra, 2026); our main method instead operates post-training on the additive symmetry specific to the output head, searches a scalar family, and selects by the prediction fidelity of the quantized frozen model. Our scalar search uses the additive equivalence of a linear-softmax output head and leaves its input representation unchanged.

Softmax invariance and output centering. SageAttention applies fixed mean-centering to sequence-dependent key activations (Zhang et al., 2024); we instead select the magnitude of a static output-weight shift according to post-quantization predictive fidelity. Output Embedding Centering uses the vocabulary-row mean to stabilize language-model pretraining (Stollenwerk et al., 2026). Consequently, neither the softmax symmetry nor mean-centering is a new contribution here. Table 2 quantifies how much the selected amount adds over fixed centering across heads and quantizers.

Reconstruction and distributional objectives. AdaRound, BRECQ, Optimal Brain Compression and GPTQ develop reconstruction objectives and compensation procedures for post-training quantization (Nagel et al., 2020; Li et al., 2021; Frantar et al., 2022; 2023). Importance-matrix implementations provide diagonal activation-weighted fitting (ggml-org, 2026; vLLM Project, 2026). Concurrent SoftWater uses feature covariance and softmax curvature for class-aware quantization rate allocation (Cavalcanti & Wilson, 2026). We use curvature as a diagnostic and measured KL to select a representative before applying a base quantizer.

Large-vocabulary prediction and execution. Adaptive softmax changes the organization of large-vocabulary prediction (Grave et al., 2017). ARCHead compresses output heads with low-rank structure and an INT4 residual correction (Kocabay et al., 2026); our intervention instead keeps the dense head and only reparameterizes it before an existing quantizer. Dense head quantization retains all vocabulary outputs. Tied-weight separation and packed Marlin/vLLM execution are established deployment tools (Kurtic et al., 2023; IST-DASLab, 2025; Kwon et al., 2023); our serving measurements quantify their relevance to output-head compression.

6 CONCLUSION

Functionally equivalent output heads can exhibit markedly different low-bit behavior. At W4, gains concentrate on heads with substantial baseline quantization error; at W2, where distortion increases across all evaluated heads, the benefit broadens across nearly the full model–quantizer matrix. The gains persist across three quantizers and stronger calibration, scaling and affine-quantization controls. Separate holdouts confirm the improvements on Phi and BLOOM. Matched residual analysis shows that a better representative need not reduce reconstruction error, but can place that error in directions that matter less to the predictive distribution. Packed inference confirms that this fidelity recovery preserves the W4 latency benefit. More broadly, an output head’s apparent precision requirement can depend on its parameterization, not only on the function it represents.

REPRODUCIBILITY STATEMENT

Appendix A specifies the data partitions, coefficient grid, selection rule, base quantizers, and numerical conventions; Appendix H documents model checkpoints and head dimensions. Appendices C–G and J provide experiment-specific protocols for holdout and transfer evaluations, stronger quantization controls, mechanism analysis, packed inference, and grouped search. These include sample counts, uncertainty estimation, validation-subsampling seeds, and deployment hardware and timing procedures. Upon publication, we will release the quantization and evaluation code, machine-readable experiment records, and scripts for regenerating the principal result tables.

REFERENCES

- Saleh Ashkboos, Amirkeivan Mohtashami, Maximilian L. Croci, Bo Li, Pashmina Cameron, Martin Jaggi, Dan Alistarh, Torsten Hoefer, and James Hensman. QuaRot: Outlier-free 4-bit inference in rotated LLMs. *arXiv preprint arXiv:2404.00456*, 2024. URL <https://arxiv.org/abs/2404.00456>.
- Miguel P. Bento and João F. Seabra. GaugeQuant: Online learning of quantization-optimal bases from LLM symmetries, 2026. URL <https://arxiv.org/abs/2607.20757>.
- Joao V. Cavalcanti and Ashia C. Wilson. SoftWater: Class-aware rate allocation for softmax quantization, 2026. URL <https://arxiv.org/abs/2608.12026>.
- Elias Frantar, Sidak Pal Singh, and Dan Alistarh. Optimal brain compression: A framework for accurate post-training quantization and pruning. In *Advances in Neural Information Processing Systems*, 2022. URL <https://arxiv.org/abs/2208.11580>.
- Elias Frantar, Saleh Ashkboos, Torsten Hoefer, and Dan Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2210.17323>.
- ggml-org. llama.cpp importance matrix documentation, 2026. URL <https://github.com/ggml-org/llama.cpp/blob/master/tools/imatrix/README.md>.
- Google. Gemma 4 E4B model configuration, 2026. URL <https://huggingface.co/google/gemma-4-E4B-it/blob/main/config.json>.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL <https://arxiv.org/abs/2407.21783>.
- Édouard Grave, Armand Joulin, Moustapha Cissé, David Grangier, and Hervé Jégou. Efficient softmax approximation for GPUs. In *International Conference on Machine Learning*, pp. 1302–1310, 2017. URL <https://proceedings.mlr.press/v70/grave17a.html>.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pp. 1321–1330, 2017. URL <https://proceedings.mlr.press/v70/guo17a.html>.

- Hongzhi Huang, Defa Zhu, Banggu Wu, Yutao Zeng, Ya Wang, Qiyang Min, and Xun Zhou. Over-tokenized transformer: Vocabulary is generally worth scaling. In *International Conference on Machine Learning (ICML)*, 2025. URL <https://arxiv.org/abs/2501.16975>.
- IST-DASLab. MARLIN: Mixed-precision auto-regressive parallel inference on large language models, artifact, 2025. URL https://github.com/IST-DASLab/marlin_artifact.
- Şuayp Talha Kocabay, Talha Rüzgar Akkuş, and Kamer Ali Yuksel. ARCHHead: Activation-metric residual correction for large language model output heads, 2026. URL <https://arxiv.org/abs/2608.02703>.
- Eldar Kurtic, Denis Kuznedelev, Elias Frantar, Michael Goin, and Dan Alistarh. Sparse fine-tuning for inference acceleration of large language models, 2023. URL https://neurips2023-enlsp.github.io/papers/paper_47.pdf.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *ACM Symposium on Operating Systems Principles*, 2023. URL <https://arxiv.org/abs/2309.06180>.
- Yuhang Li, Ruihao Gong, Xu Tan, Yang Yang, Peng Hu, Qi Zhang, Fengwei Yu, Wei Wang, and Shi Gu. BRECQ: Pushing the limit of post-training quantization by block reconstruction. In *International Conference on Learning Representations*, 2021. URL <https://arxiv.org/abs/2102.05426>.
- Ji Lin, Jiaming Tang, Haotian Tang, et al. AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration. In *Proceedings of Machine Learning and Systems*, 2024. URL https://proceedings.mlsys.org/paper_files/paper/2024/hash/42a452cbafa9dd64e9ba4aa95cc1ef21-Abstract-Conference.html.
- Zechun Liu, Changsheng Zhao, Igor Fedorov, Bilge Soran, Dhruv Choudhary, Raghuraman Krishnamoorthi, Vikas Chandra, Yuandong Tian, and Tijmen Blankevoort. SpinQuant: LLM quantization with learned rotations. In *International Conference on Learning Representations*, 2025. URL <https://arxiv.org/abs/2405.16406>.
- James Martens. New insights and perspectives on the natural gradient method. *Journal of Machine Learning Research*, 21(146):1–76, 2020. URL <https://jmlr.org/papers/v21/17-678.html>.
- Microsoft. Phi-4-mini-instruct model configuration, 2025. URL <https://huggingface.co/microsoft/Phi-4-mini-instruct/blob/main/config.json>.
- Mistral AI. Ministral-3-3B-Instruct-2512-BF16 model configuration, 2025. URL <https://huggingface.co/mistralai/Ministral-3-3B-Instruct-2512-BF16/blob/main/config.json>.
- Markus Nagel, Rana Ali Amjad, Mart van Baalen, Christos Louizos, and Tijmen Blankevoort. Up or down? Adaptive rounding for post-training quantization. In *International Conference on Machine Learning*, pp. 7197–7206, 2020. URL <https://proceedings.mlr.press/v119/nagel20a.html>.
- Qwen Team. Qwen3.5-4B model configuration, 2026. URL <https://huggingface.co/Qwen/Qwen3.5-4B/blob/main/config.json>.
- Qwen Team, Alibaba Cloud. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. URL <https://arxiv.org/abs/2412.15115>.
- Felix Stollenwerk, Anna Lokrantz, and Niclas Hertzberg. Output embedding centering for stable LLM pretraining, 2026. URL <https://arxiv.org/abs/2601.02031>.
- Chaofan Tao, Qian Liu, Longxu Dou, Niklas Muennighoff, Zhongwei Wan, Ping Luo, Min Lin, and Ngai Wong. Scaling laws with vocabulary: Larger models deserve larger vocabularies. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL <https://arxiv.org/abs/2407.13623>.

vLLM Project. iMatrix importance-weighted quantization, 2026. URL <https://docs.vllm.ai/projects/llm-compressor/en/latest/examples/imatrix/>. Documentation accessed September 5, 2026.

Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. SmoothQuant: Accurate and efficient post-training quantization for large language models. In *International Conference on Machine Learning*, 2023. URL <https://arxiv.org/abs/2211.10438>.

Jintao Zhang, Jia Wei, Pengle Zhang, Jun Zhu, and Jianfei Chen. SageAttention: Accurate 8-bit attention for plug-and-play inference acceleration, 2024. URL <https://arxiv.org/abs/2410.02367v1>.

APPENDIX

The supplement begins with the experimental protocol, search pseudocode and base-quantizer conventions (Appendix A), followed by the nonlinear extension (Appendix B). Appendices C–F report additional evaluations, mean-centering controls and residual diagnostics. The frozen English-selected FLORES transfer check is in Appendix C.2; the C4/OpenWebMath transfer matrix and BLOOM coefficient-stability study are in Appendix C.3. Appendix D gives the Phi calibration, exact-scaling, affine-quantization and direction controls of Section 3.2. Appendix G documents packed deployment and storage; Appendix H supplies model dimensions and the checkpoint sources for Figure 1; Appendix I reports confidence controls. Appendix J describes the higher-capacity grouped parameterization, nine-model results and a separate SmolLM3 comparison with matched candidate counts. Numerical paths and BF16 references are matched within each comparison; results from different protocols are not pooled.

A EXPERIMENTAL PROTOCOL AND BASE QUANTIZERS

A.1 DATA, SELECTION AND EVALUATION

The matrix and breadth probes capture final hidden states from frozen BF16 models. The pinned WikiText article-selection routine assigns 128 training articles to fitting, with eight positions per article (1,024 states), and partitions 32 test articles into 16 validation and 16 evaluation articles. Prefixes contain at most 512 tokens. Fitting estimates activation moments; validation selects the coefficient; evaluation reports KL, top-1 agreement and perplexity. Linear-logit products and scoring use FP32. Gemma 4 instead applies its soft-cap after the correction in Appendix B.

The ordered candidate grid is

$$\mathcal{T} = (-2, -1, -0.5, 0, 0.5, 1, 1.5, 2, 2.5, 3, 4, 5, 6, 8).$$

The grid includes the original head ($t = 0$) and fixed mean-centering ($t = 1$), with finer spacing near these baselines, coarser spacing at larger positive shifts, and negative candidates to test the opposite direction. The same grid is used for every model and base quantizer in the matrix and breadth evaluations. Each base quantizer selects its own coefficient from unrounded validation KL. The recompute implementations retain the first candidate on an exact tie. They reuse fitting moments across candidates: AW-MSE uses diagonal moments, and GPTQ reuses the same fitting Hessian while quantizing each representative independently. The four-head RTN/AW-MSE search takes about five minutes on one A100, and Phi’s 14-point GPTQ sweep about twelve.

Although the three article sets are disjoint within a run, the diagnostic pool was used in earlier pilot exploration. The separate 26-article untouched holdout is reported in Appendix C.1. A plotted evaluation curve is post-selection analysis; it does not provide a new selection set. Small changes in the diagnostic matrix lack per-article uncertainty. Packed deployment uses a separate block-based protocol and source reference (Appendix G.1).

Algorithm 1 Validation-selected softmax reparameterization

Require: Source head W , logit path g , base quantizer $\mathcal{B}$;
fitting states $\mathcal{H}_{\text{fit}}$, validation states $\mathcal{H}_{\text{val}}$;
fixed ordered grid $\mathcal{T}$ containing 0

- 1: $\mu \leftarrow W^\top \mathbf{1}/V$; estimate quantizer moments from $\mathcal{H}_{\text{fit}}$
- 2: Set $r = 0$ if g preserves common shifts, otherwise $r = 1$
- 3: Verify unquantized equivalence for all $t \in \mathcal{T}$, correcting if $r = 1$
- 4: $L^* \leftarrow +\infty$
- 5: **for** t in $\mathcal{T}$, in order **do**
- 6: $W_t \leftarrow W - t\mathbf{1}\mu^\top$; $Q_t \leftarrow \mathcal{B}(W_t; \mathcal{H}_{\text{fit}})$
- 7: $p(h) \leftarrow \text{softmax}(g(W_t h))$
- 8: $q_t(h) \leftarrow \text{softmax}(g(Q_t h + r t(\mu^\top h)\mathbf{1}))$
- 9: $L_t \leftarrow |\mathcal{H}_{\text{val}}|^{-1} \sum_{h \in \mathcal{H}_{\text{val}}} D_{\text{KL}}(p(h) \| q_t(h))$
- 10: **if** $L_t < L^*$ **then**
- 11: $(L^*, t^*, Q^*) \leftarrow (L_t, t, Q_t)$
- 12: **end if**
- 13: **end for**
- 14: **return** Q^* ; retain (t^*, μ) for inference only if $r = 1$

Algorithm 1 uses the actual model logit path; an unchanged bias can be included in g . Final evaluation states are not inputs to the search. Quantizer precision, groups, clipping grid and numerical conventions remain fixed across candidates. The equivalence check permits numerical roundoff as described in Appendix B.

A.2 BASE QUANTIZERS

Let $k_b = 2^{b-1} - 1$. For a row group, the candidate scale is $s_g(c) = c \max_{j \in g} |W_{vj}|/k_b$, with a positive numerical floor for zero ranges. Codes round $W_{vj}/s_g(c)$ and clip to the integer range specified in Table 5. The table also specifies whether scales are cast to BF16 before candidate scoring or only when stored.

RTN fixes $c = 1$. AW-MSE uses $m_j = \mathbb{E}_{\text{fit}}[h_j^2]$ and selects

$$c_g^* = \arg \min_{c \in \mathcal{C}} \sum_{j \in g} m_j (W_{vj} - Q_{vj}(c))^2. \quad (7)$$

This is an established importance-weighted primitive (ggml-org, 2026; vLLM Project, 2026); unweighted MSE sets $m_j = 1$. GPTQ uses AW-MSE initial scales and propagates rounding error with the full fitting second moment $H = 2X^\top X/N$, 1% damping, block size 128 and no activation ordering (Frantar et al., 2023). All heads use G128 unless noted.

A.3 NUMERICAL CONVENTIONS BY EXPERIMENT

For the matrix AW-MSE/GPTQ and matched probability probes, the clipping grid is

$$\mathcal{C} = (1, .975, .95, .925, .9, .875, .85, .8, .75, .7, .6, .5).$$

The packed Phi AW-MSE path uses the first ten factors, ending at 0.7. RTN uses only $c = 1$. A raw/shifted pair always shares its convention; cross-quantizer comparisons can also differ in scale precision and integer range.

The reconstruction scan in Appendix E.2 is a separate symmetric-RTN diagnostic with its own shorter coefficient grid. Integrated likelihood measurements use the serving model’s logit path and BF16 reference; probability probes evaluate the specified logit transform and softmax directly, without deployment sampling penalties.

Implementation and evidence provenance. Packing, tied-weight separation and kernel verification are documented in Appendix G. The measured packed Phi path is shift-compatible and requires no rank-one correction. Gemma 4’s correction is evaluated in the dense probability probes; its packed deployment was not measured. The released build script regenerates the matrix, breadth and transfer

Table 5: Numerical conventions for the reported comparisons. “Signed” is $[-2^{b-1}, k_b]$; “symmetric” is $[-k_b, k_b]$. Reconstruction precision describes the weights presented to the scoring path.

Comparison	Codes	Scale selection/storage	Reconstruction
Matrix/breadth RTN	Symmetric	FP32	FP32
Matrix/breadth AW-MSE	Signed	Score and store BF16	BF16, then FP32 products
GPTQ probe	Signed	BF16 initial scales	BF16, then FP32 products
Matched Phi residuals	Signed	Score and store BF16	BF16, then FP32 products
Packed Phi RTN/AW-MSE	Signed	Float candidates; store BF16	Packed W4 serving

tables from per-cell records, including record hashes, selected coefficients and the transfer-regression check. Captured-state hashes and checkpoint revisions are also needed for bit-exact reproduction; limitations of the exploratory scan are stated in Appendix E.2.

B NONLINEAR LOGIT PATHS AND INFERENCE COMPATIBILITY

Some models apply a nonlinear transformation to logits before softmax. A common shift before that transformation can become unequal changes afterward, altering the predictions. For these models, we restore the removed shared component before applying the nonlinearity.

B.1 SHIFT-COMPATIBLE LOGIT PATHS

Equation 1 assumes the complete pre-softmax logit path preserves vocabulary-wide additive shifts. Let g denote every operation applied to the linear logits before softmax. The reparameterization is exact precisely when

$$\text{softmax}(g(z + c\mathbf{1})) = \text{softmax}(g(z)) \quad \text{for all } z, c, \quad (8)$$

Equivalently, $P g(z + c\mathbf{1}) = P g(z)$ with $P = I - \mathbf{1}\mathbf{1}^\top / V$. Identity logits, temperature scaling and an unchanged additive bias satisfy this; an elementwise nonlinearity such as tanh logit soft-capping does not, because $s \tanh((z + c\mathbf{1})/s)$ is not a common shift of $s \tanh(z/s)$.

B.2 RANK-ONE CORRECTION

When g violates Equation 8, the shared component can be retained exactly rather than discarded. Decompose the head as $W = (W - t\mathbf{1}\mu^\top) + t\mathbf{1}\mu^\top$, quantize only the first term to Q_t , and restore the rank-one term to the quantized logits *before* g :

$$z_t = Q_t h + t(\mu^\top h)\mathbf{1}, \quad p_t = \text{softmax}(g(z_t)). \quad (9)$$

With $Q_t = W_t$ this gives $z_t = Wh$ exactly, so $p_t = p_W$ for *any* g ; on Gemma-4 the unquantized shifted head reproduces the source distribution to $\text{KL} \approx 7 \times 10^{-10}$. After quantization, the only residual entering g is $E_t h$, where $E_t = Q_t - W_t$. The correction costs one d -dimensional dot product and a scalar broadcast over the vocabulary, compared with Vd multiply-accumulates for the projection, and can be fused into a soft-capping pass; for a pure softmax path it may be omitted because softmax discards $\mathbf{1}$. We still execute each shifted head through the model’s actual logit path and verify $p_t = p_W$ before searching t , since casting to BF16 introduces roundoff and the check is empirical rather than a claim of bitwise equivalence.

Storage and tied weights. The correction retains t and the d -dimensional row mean μ , an $O(d)$ overhead relative to the $O(Vd)$ head. For tied weights, the input embedding retains its source values while a separate output copy is quantized, using established weight separation (Kurtic et al., 2023). Appendix G.2 gives the resulting storage accounting.

The Gemma 4 equivalence and fidelity measurements use dense probability probes. The packed timing results in Appendix G.1 concern Phi and do not measure the runtime cost of this correction.

C EXTENDED ROBUSTNESS AND GENERALIZATION

This section reports an untouched-data replication in two independent model families, multilingual and cross-domain transfer of frozen English-selected coefficients, validation-size stability, and the extension to lower precisions.

C.1 UNTOUCHED HOLDOUT REPLICATION

Section 3 reports the Phi and BLOOM results on the 26-article untouched holdout (13,286 tokens). Article-bootstrap 95% intervals use 10,000 replicates: Phi KL reductions are [72.9, 74.4]% under AW-MSE and [70.4, 71.7]% under RTN; BLOOM intervals are [72.4, 76.2]% and [48.7, 54.3]%, respectively. The per-cell records accompany the code release. Re-running the frozen rule on the original evaluation split reproduces the Phi matrix values exactly (RTN 1.229 to 0.351 KL, AW-MSE 0.936 to 0.256; AW-MSE PPL 24.72 to 12.50, BF16 9.73). This checks implementation consistency; the separate holdout supplies the generalization test.

Table 6 retains the likelihood measurements and exact KL pairs. BLOOM GPTQ improves by 23.4% using unrounded KL values; no bootstrap interval is reported for that comparison.

Table 6: **KL reductions persist on untouched holdouts.** Pairs are original $\rightarrow$ shifted; BF16 PPL uses the same source and evaluation path. Dashes denote unreported values.

Head	Quantizer	KL	PPL	BF16 PPL
Phi	AW-MSE	0.943 $\rightarrow$ 0.249	23.41 $\rightarrow$ 11.93	9.33
Phi	RTN	1.204 $\rightarrow$ 0.348	31.00 $\rightarrow$ 12.93	9.33
BLOOM	AW-MSE	0.558 $\rightarrow$ 0.142	—	—
BLOOM	GPTQ	0.037 $\rightarrow$ 0.028	—	—

C.2 FROZEN ENGLISH-SELECTED SHIFTS ON FLORES

We evaluate the existing English WikiText-selected W4 G128 heads on the first 256 aligned sentences of the original FLORES-200 devtest release, independently in each of six languages, without refitting or retuning t .¹ The BF16 decoder is frozen; each sentence supplies at most 128 tokens (none were truncated), and Table 7 reports token-weighted, full-vocabulary $D_{\text{KL}}(p_W \| p_Q)$ on all next-token positions, using FP32 readouts and FP64 reductions with TF32 disabled. XGLM improves in all 18 language-quantizer point estimates, with 17 paired intervals excluding zero. BLOOM instead increases KL by 67.3% for Arabic/RTN, 13.4% for Hindi/RTN, and 82.7% for Hindi/AW-MSE; all three intervals exclude zero. Its small Chinese/GPTQ increase is uncertain. Both models improve under all three quantizers on English FLORES. These results expose distribution-dependent transfer of frozen heads; they neither guarantee multilingual transfer nor establish that the residual moved specifically into non-English token coordinates. This is a short-context fidelity diagnostic, not a translation-quality evaluation.

C.3 CROSS-DOMAIN TRANSFER AND COEFFICIENT STABILITY

Frozen WikiText coefficients. We evaluate four heads under W4 G128 RTN, AW-MSE and full-Hessian GPTQ on C4 English and OpenWebMath. Let t_{WT} denote the coefficient selected on the historical WikiText validation set and frozen before this evaluation. Each domain supplies 256 documents, with all next-token positions scored in prefixes of at most 512 tokens. The decoder remains BF16; readouts use FP32. No target-domain data are used to fit the quantizer or select t_{WT} . GPTQ uses a 65,536-state Hessian and the original 1,024-state range initializer. Consequently, BLOOM’s frozen GPTQ coefficient is 5, whereas the original 1,024-state configuration in Table 1 selects 4.

Table 8 reports all 24 comparisons. The frozen coefficient improves over mean-centering in all 18 cases where $t_{\text{WT}} \neq 1$; all corresponding paired 95% intervals exclude zero. XGLM ties mean-centering in six cases because $t_{\text{WT}} = 1$. All 24 frozen-coefficient KL point estimates are below the

¹Meta’s original FLORES-200 release.

Table 7: **English-selected shifts lower all 18 XGLM FLORES KL values but can regress on BLOOM.** Entries are raw $\rightarrow$ shifted KL (nats). BLOOM uses $t = (6, 4, 4)$ for (RTN, AW-MSE, GPTQ); XGLM uses $t = 1$. GPTQ retains the original 1,024-state Hessian and ranges. Bold marks increased KL; $\dagger$ marks paired 95% intervals containing zero: 5,000 bootstrap resamples of 75 source-article URL clusters, without multiplicity adjustment. Arabic is Modern Standard Arabic; Chinese uses Simplified script.

Model	Language	RTN	AW-MSE	GPTQ
BLOOM-1.7B	English	1.586599 $\rightarrow$ 0.665851	0.920758 $\rightarrow$ 0.144382	0.055268 $\rightarrow$ 0.038390
	French	0.946322 $\rightarrow$ 0.556638	0.845735 $\rightarrow$ 0.208255	0.713031 $\rightarrow$ 0.250820
	Spanish	2.074753 $\rightarrow$ 0.762232	1.565073 $\rightarrow$ 0.273351	0.815738 $\rightarrow$ 0.297205
	Arabic	0.523881 $\rightarrow$ 0.876482	0.150436 $\rightarrow$ 0.148087 †	0.236476 $\rightarrow$ 0.209441
	Hindi	1.629840 $\rightarrow$ 1.847837	0.230221 $\rightarrow$ 0.420559	0.951959 $\rightarrow$ 0.447360
	Chinese	0.641800 $\rightarrow$ 0.625781	0.083138 $\rightarrow$ 0.067635	0.118631 $\rightarrow$ 0.119211†
XGLM-1.7B	English	1.849360 $\rightarrow$ 0.167656	0.427737 $\rightarrow$ 0.104735	0.011856 $\rightarrow$ 0.008691
	French	1.250451 $\rightarrow$ 0.168155	0.214890 $\rightarrow$ 0.106347	0.022430 $\rightarrow$ 0.017270
	Spanish	1.040484 $\rightarrow$ 0.169850	0.229209 $\rightarrow$ 0.117290	0.023801 $\rightarrow$ 0.019992
	Arabic	0.658988 $\rightarrow$ 0.235897	0.189734 $\rightarrow$ 0.171468	0.052101 $\rightarrow$ 0.051720 †
	Hindi	1.249754 $\rightarrow$ 0.254091	0.263036 $\rightarrow$ 0.163331	0.058555 $\rightarrow$ 0.048005
	Chinese	1.610301 $\rightarrow$ 0.227931	0.312039 $\rightarrow$ 0.155611	0.031372 $\rightarrow$ 0.029112

unshifted baseline. Qwen3.5 illustrates that fixed mean-centering can worsen fidelity: $t = 1$ increases KL in every cell, while the frozen $t_{WT} = -0.5$ improves it.

Table 8: **Frozen WikiText coefficients beat centering in 18 transfer comparisons and tie in six.** C4/OpenWebMath test KL to BF16 (nats): unshifted ($t = 0$), centered ($t = 1$), and frozen ($t = t_{WT}$). Frozen-minus-centered KL intervals use 10,000 paired document-bootstrap resamples; they are pointwise, conditional on frozen coefficients, and unadjusted for multiple comparisons.

Head	Quantizer	Domain	t_{WT}	Raw KL	Centered KL	Frozen KL	Δ KL: 95% CI
Phi-4-mini	RTN	C4	4	1.372879	0.899519	0.393513	$[-0.517935, -0.493749]$
Phi-4-mini	RTN	OpenWebMath	4	1.105418	0.675704	0.342138	$[-0.348787, -0.318388]$
Phi-4-mini	AW-MSE	C4	4	1.039478	0.657195	0.304324	$[-0.359930, -0.345035]$
Phi-4-mini	AW-MSE	OpenWebMath	4	0.764193	0.498817	0.252586	$[-0.256702, -0.235430]$
Phi-4-mini	GPTQ	C4	4	0.111289	0.078885	0.041984	$[-0.037799, -0.035971]$
Phi-4-mini	GPTQ	OpenWebMath	4	0.101269	0.072217	0.044054	$[-0.029566, -0.026734]$
BLOOM-1.7B	RTN	C4	6	1.576803	1.226448	0.585626	$[-0.669812, -0.611228]$
BLOOM-1.7B	RTN	OpenWebMath	6	1.055612	0.872942	0.526070	$[-0.371997, -0.322349]$
BLOOM-1.7B	AW-MSE	C4	4	0.943439	0.691491	0.147667	$[-0.565035, -0.522129]$
BLOOM-1.7B	AW-MSE	OpenWebMath	4	0.479431	0.396989	0.124050	$[-0.293043, -0.253344]$
BLOOM-1.7B	GPTQ	C4	5	0.043011	0.044096	0.033579	$[-0.011303, -0.009735]$
BLOOM-1.7B	GPTQ	OpenWebMath	5	0.079911	0.071089	0.057823	$[-0.014991, -0.011484]$
XGLM-1.7B	RTN	C4	1	1.833599	0.154811	0.154811	$[+0.000000, +0.000000]$
XGLM-1.7B	RTN	OpenWebMath	1	2.387851	0.159820	0.159820	$[+0.000000, +0.000000]$
XGLM-1.7B	AW-MSE	C4	1	0.478122	0.098329	0.098329	$[+0.000000, +0.000000]$
XGLM-1.7B	AW-MSE	OpenWebMath	1	0.733782	0.095567	0.095567	$[+0.000000, +0.000000]$
XGLM-1.7B	GPTQ	C4	1	0.009799	0.007307	0.007307	$[+0.000000, +0.000000]$
XGLM-1.7B	GPTQ	OpenWebMath	1	0.009308	0.006844	0.006844	$[+0.000000, +0.000000]$
Qwen3.5	RTN	C4	-0.5	0.029287	0.041919	0.022992	$[-0.019664, -0.018170]$
Qwen3.5	RTN	OpenWebMath	-0.5	0.020447	0.029660	0.016817	$[-0.013380, -0.012298]$
Qwen3.5	AW-MSE	C4	-0.5	0.015426	0.025728	0.014890	$[-0.011288, -0.010381]$
Qwen3.5	AW-MSE	OpenWebMath	-0.5	0.012674	0.019580	0.011050	$[-0.008928, -0.008129]$
Qwen3.5	GPTQ	C4	-0.5	0.008057	0.011161	0.007906	$[-0.003389, -0.003122]$
Qwen3.5	GPTQ	OpenWebMath	-0.5	0.008031	0.011584	0.007309	$[-0.004458, -0.004098]$

Coefficient selection and validation size on BLOOM. A separate study evaluates all 14 coefficients on BLOOM, holding quantizer fitting fixed. Each domain has 2,048 validation and 1,024 disjoint test documents, with 32 sampled next-token positions per prefix. Table 9 compares historical WikiText coefficients with those selected on the full C4 and OpenWebMath validation pools. These are finite-grid choices under unequal selection budgets, not continuous or population optima.

Table 9: **BLOOM’s selected coefficient varies across validation domains.** Column headings give validation sizes in articles/documents. WikiText coefficients are historical; C4 and OpenWebMath selections use separate 2,048-document pools.

Quantizer	WikiText (16)	C4 (2,048)	OpenWebMath (2,048)
RTN	6	4	4
AW-MSE	4	4	5
GPTQ	5	4	4

We repeat selection using seven nested subset sizes (16–1,024 documents) and ten seeds per domain, freezing each choice before test scoring. Table 10 summarizes the smallest and largest sizes. At 1,024 documents, all ten seeds agree for each of the six domain–quantizer pairs, selecting $t = 4$ or $t = 5$. Several pairs already agree at 16 documents; others vary among nearby grid candidates. The reported test KL averages the ten selected heads’ losses on the same fixed test set. This supports stable non-unit choices for these two domains, without establishing universal stability: in the broader five-language mC4 study, French RTN still selects $t = 4$ in three seeds and $t = 5$ in seven at 1,024 documents.

Table 10: **At 1,024 validation documents, all ten seeds agree on a non-unit BLOOM coefficient per setting.** Counts show selection frequencies. Test KL averages seeds on a fixed, disjoint 1,024-document set unused for selection.

Quantizer	Domain	Selection at $n = 16$	Selection at $n = 1,024$	Test KL: 16 $\rightarrow$ 1,024
RTN	C4	$t = 4: 6/10; t = 6: 4/10$	$t = 4: 10/10$	0.572704 $\rightarrow$ 0.563011
RTN	OpenWebMath	$t = 4: 9/10; t = 6: 1/10$	$t = 4: 10/10$	0.532576 $\rightarrow$ 0.531784
AW-MSE	C4	$t = 4: 10/10$	$t = 4: 10/10$	0.146926 $\rightarrow$ 0.146926
AW-MSE	OpenWebMath	$t = 4: 3/10; t = 5: 7/10$	$t = 5: 10/10$	0.126947 $\rightarrow$ 0.126493
GPTQ	C4	$t = 3: 3/10; t = 4: 7/10$	$t = 4: 10/10$	0.033628 $\rightarrow$ 0.033356
GPTQ	OpenWebMath	$t = 3: 5/10; t = 4: 5/10$	$t = 4: 10/10$	0.054443 $\rightarrow$ 0.053538

Protocol distinctions. The transfer and stability studies use different document sets and position sampling; differences in their absolute KL values should not be attributed solely to coefficient selection. After temporary head banks were lost, AW-MSE and GPTQ heads were rebuilt on A100 from pinned source weights and archived calibration document/token identities. Their tensors differ from the historical heads; RTN tensors match. The transfer experiment therefore freezes coefficients, not every historical quantized tensor. Comparisons within each study use matched reconstructed candidates. Historical versus new-domain selection also differs in validation-set size, so these experiments do not isolate distribution shift from selection-budget and reconstruction effects.

C.4 LOWER-PRECISION EXTENSION (W3 AND W2)

Table 11 extends the seven-head comparison to W3 and W2 using the protocol in Appendix A.1. At W3, Phi test KL falls from 12.97 to 2.46 under RTN, 4.62 to 1.12 under AW-MSE, and 0.543 to 0.211 under GPTQ. W2 remains substantially degraded even with the shift: Phi GPTQ reaches PPL 30.0 versus the matched BF16 reference of 9.73, while Gemma 3 RTN falls from 248.3 to 146.3. The relative KL reduction is largest for RTN and smallest for GPTQ at W3; this ordering does not persist at W2. No head’s test KL increases in this comparison.

Table 12 gives W3 perplexities from FP32 readouts on the same 8,176-position evaluation split. Its BF16 column is the unquantized source head on the same states. Very large RTN values reflect near-degenerate heads: Phi falls from 3.98×10^6 to 111.5 with the shift, and XGLM from 1.5×10^{14} to 34.5 (BF16 13.8). Phi GPTQ reaches 11.97, compared with BF16 9.73.

Table 11: **Shifting lowers or preserves KL at W3 and W2, but W2 distortion remains large.** Each cell is raw ($t = 0$) $\rightarrow$ reparameterized (t^*) test KL to the source, with per-quantizer validation selection over the same 14-point grid. The four original heads appear above each rule; three very-large-vocabulary heads appear below. Gemma 4 ($\dagger$) uses the rank-one soft-cap correction.

Bits	Model	RTN	AW-MSE	GPTQ
W2	Phi	98.2 $\rightarrow$ 74.2	41.6 $\rightarrow$ 16.2	3.08 $\rightarrow$ 1.16
W2	Gemma 3	2.66 $\rightarrow$ 2.21	0.822 $\rightarrow$ 0.714	0.508 $\rightarrow$ 0.491
W2	Gemma 4 $\dagger$	0.911 $\rightarrow$ 0.911	0.212 $\rightarrow$ 0.199	0.165 $\rightarrow$ 0.162
W2	Qwen3.5	1.84 $\rightarrow$ 1.74	0.312 $\rightarrow$ 0.285	0.2 $\rightarrow$ 0.197
W2	BLOOM-1.7B	16.7 $\rightarrow$ 6.15	4.46 $\rightarrow$ 1.46	0.427 $\rightarrow$ 0.335
W2	BLOOMZ-1.7B	16.2 $\rightarrow$ 6.22	3.99 $\rightarrow$ 2.12	0.456 $\rightarrow$ 0.367
W2	XGLM-1.7B	97.8 $\rightarrow$ 55.3	23.7 $\rightarrow$ 3.73	0.173 $\rightarrow$ 0.128
W3	Phi	13 $\rightarrow$ 2.46	4.62 $\rightarrow$ 1.12	0.543 $\rightarrow$ 0.211
W3	Gemma 3	0.244 $\rightarrow$ 0.244	0.156 $\rightarrow$ 0.144	0.112 $\rightarrow$ 0.112
W3	Gemma 4 $\dagger$	0.0655 $\rightarrow$ 0.062	0.0343 $\rightarrow$ 0.0337	0.0352 $\rightarrow$ 0.0341
W3	Qwen3.5	0.105 $\rightarrow$ 0.0971	0.0566 $\rightarrow$ 0.0523	0.0419 $\rightarrow$ 0.0403
W3	BLOOM-1.7B	2.7 $\rightarrow$ 1.72	1.02 $\rightarrow$ 0.443	0.0969 $\rightarrow$ 0.0839
W3	BLOOMZ-1.7B	2.6 $\rightarrow$ 1.58	0.943 $\rightarrow$ 0.46	0.107 $\rightarrow$ 0.0983
W3	XGLM-1.7B	30 $\rightarrow$ 0.926	14.2 $\rightarrow$ 0.372	0.0352 $\rightarrow$ 0.0265

Table 12: **KL-selected shifts can substantially reduce W3 perplexity, but gains are not uniform.** Each quantizer cell is raw ($t = 0$) $\rightarrow$ reparameterized (t^*) perplexity. BF16 is the unquantized source head on the same 8,176-position test split (FP32 readout). Gemma 4 ($\dagger$) uses the rank-one soft-cap correction.

Head	BF16	RTN	AW-MSE	GPTQ
Phi	9.73	3.98 $\times 10^6 \rightarrow$ 111.5	1002 $\rightarrow$ 28.47	16.86 $\rightarrow$ 11.97
Gemma 3	38.49	44.34 $\rightarrow$ 44.34	38.54 $\rightarrow$ 38.89	40.09 $\rightarrow$ 40.09
Gemma 4 $\dagger$	66.41	71.11 $\rightarrow$ 70.83	68.71 $\rightarrow$ 69.06	69.05 $\rightarrow$ 68.85
Qwen3.5	8.98	9.851 $\rightarrow$ 9.793	9.503 $\rightarrow$ 9.446	9.351 $\rightarrow$ 9.291
BLOOM-1.7B	18.47	282.3 $\rightarrow$ 100.7	50.78 $\rightarrow$ 26.89	20.14 $\rightarrow$ 20.04
BLOOMZ-1.7B	22.06	285.3 $\rightarrow$ 124.9	56.08 $\rightarrow$ 31.42	23.8 $\rightarrow$ 23.9
XGLM-1.7B	13.80	1.53 $\times 10^{14} \rightarrow$ 34.46	2.12 $\times 10^7 \rightarrow$ 20.27	14.19 $\rightarrow$ 14.16

D COMPLEMENTARITY AND STRONGER CONTROLS ON PHI

Protocol and scope. We recompute the controls on Phi-4-mini at W4/G128 with a frozen BF16 decoder, 128 fitting articles, 16 validation articles and 16 disjoint test articles, using 512-token article prefixes and the same 14-point scalar grid as Appendix A. These are the original evaluation articles, not the untouched holdout of Appendix C.1. All choices are frozen on validation KL before test evaluation. Source test PPL is 9.7192. The run uses FP32 readouts with TF32 disabled, torch 2.12.0 and transformers 5.10.1; its matched baselines are recomputed rather than pooled with earlier runs. Table 13 gives the eight primary paired comparisons. They are offline quality controls, not packed-serving or latency measurements.

GPTQ calibration. The small bank contains eight states from each fitting article (1,024 total). The large bank uses all 512 prefix positions from the same articles (65,536 total) to accumulate the full second moment. We retain 1% damping, no activation ordering and the pinned llmcompressor 0.12.0.1 solver. The middle GPTQ condition changes only the Hessian while keeping the original AW-MSE range statistics; the third also refits those statistics on the large bank. All three conditions select $t = 4$. Thus the gain survives both improved error-feedback calibration and refitted initial ranges.

Exact channel scaling. For positive channel scales s , we quantize $(W - t\mathbf{1}\mu^\top)\text{diag}(s)$ and read out with $\text{diag}(s)^{-1}h$. Before quantization this preserves the source softmax. The validation

pool contains identity scaling, 20 activation-mean power candidates $s_j \propto \mathbb{E}_{\text{fit}}[|h_j|]^\alpha$ with $\alpha \in \{0.05, 0.10, \dots, 1\}$, and 11 SmoothQuant-style candidates $s_j \propto \max_{\text{fit}} |h_j|^\alpha / (\max_v |W_{vj}|)^{1-\alpha}$ with $\alpha \in \{0, 0.1, \dots, 1\}$. Scales are positive-clamped and normalized by the geometric midpoint of their minimum and maximum; activation moments for AW-MSE are transformed by s_j^{-2} . This adapts exact scaling to head-only quantization and KL selection rather than reproducing a complete AWQ or SmoothQuant pipeline. We first select scale-only, freeze that scale, and then select the additive coefficient. Both quantizers select weight-max channel equalization (the second family at $\alpha = 0$); subsequent shifts select $t = 3$ for RTN and $t = 4$ for AW-MSE. This is a sequential search, not an exhaustive scale-shift grid. Source equivalence is checked in FP64; we do not claim numerically lossless folding into a BF16 normalization or measured serving performance for the scaled variants.

Affine quantization and direction controls. Affine RTN uses group extrema including zero, a scale given by their range divided by 15, a rounded integer zero point in $[0, 15]$, and 16 reconstruction levels. Symmetric RTN retains the paper’s 15-level $[-7, 7]$ grid with FP32 scales and reconstruction; AW-MSE and GPTQ use signed $[-8, 7]$ codes, BF16 scales and BF16 reconstruction. Shift effects are compared within each quantizer. The affine condition selects $t = 5$. For direction controls, we replace μ with the coordinate-wise median or one of three fixed-seed Gaussian directions, each rescaled to $\|\mu\|_2$, and give every direction the same scalar grid (Table 14). Random shifts also help, but the mean direction outperforms all tested alternatives; these controls support its empirical utility rather than establish its optimality.

Uncertainty and evidence. Table 15 reports paired article-bootstrap intervals from 10,000 resamples (seed 20260912), recomputing token-weighted KL differences within each sampled set of articles. The intervals are descriptive and conditional on validation selection, with only 16 independent test articles. They do not establish cross-family generalization. The complete run retains 276 validation candidates, 24 distinct tested heads and 22 comparisons, including overlapping direction and combined-policy controls. Source revisions, token IDs, transform vectors, state/code hashes, frozen selections and per-article metrics are retained with the numerical implementation in the experiment records.

Table 13: **Phi shift gains persist under stronger calibration, scaling and affine quantization.** W4/G128 entries: baseline $\rightarrow$ shifted. Scaled baselines include selected channel scaling. BF16 PPL is 9.7192.

Condition	Test KL		Test PPL	KL reduction
GPTQ: 1,024 states	0.16041 $\rightarrow$ 0.06051	11.2883 $\rightarrow$ 10.2858		62.3%
GPTQ: 65,536, original ranges	0.08936 $\rightarrow$ 0.03491	10.5608 $\rightarrow$ 10.0542		60.9%
GPTQ: 65,536, refitted ranges	0.09247 $\rightarrow$ 0.03491	10.6275 $\rightarrow$ 10.0182		62.2%
RTN	1.23002 $\rightarrow$ 0.35058	34.1129 $\rightarrow$ 13.4611		71.5%
AW-MSE	0.93940 $\rightarrow$ 0.25551	24.8709 $\rightarrow$ 12.4856		72.8%
Scaled RTN	0.33659 $\rightarrow$ 0.24234	13.5410 $\rightarrow$ 12.2376		28.0%
Scaled AW-MSE	0.23998 $\rightarrow$ 0.17199	12.3055 $\rightarrow$ 11.6372		28.3%
Affine RTN	0.75002 $\rightarrow$ 0.25878	21.0794 $\rightarrow$ 12.7901		65.5%

Table 14: **The mean direction gives the lowest KL among the tested alternatives on Phi.** Median and random directions are norm-matched to the mean and use the same validation scalar search. Random shifts also improve on the raw head.

Direction	RTN			AW-MSE		
	t^*	Test KL	Test PPL	t^*	Test KL	Test PPL
Vocabulary mean	4	0.35058	13.4611	4	0.25551	12.4856
Coordinate-wise median	4	0.73863	20.1241	4	0.55894	17.0431
Random seed 0	−0.5	1.05676	27.7650	−1	0.77292	21.8672
Random seed 1	2	1.06612	28.1215	1	0.86884	22.6155
Random seed 2	−1	1.04971	27.0636	−0.5	0.84131	22.1993

Table 15: **All eight Phi KL-reduction intervals exclude zero.** Negative differences favor shifting. Paired 95% intervals use 10,000 bootstrap resamples of 16 test articles, conditional on validation selection.

Condition	Δ test KL	95% interval
GPTQ: 1,024 states	-0.099903	[-0.106154, -0.093710]
GPTQ: 65,536, original ranges	-0.054452	[-0.059731, -0.049574]
GPTQ: 65,536, refitted ranges	-0.057559	[-0.062531, -0.052901]
RTN	-0.879442	[-0.933053, -0.824182]
AW-MSE	-0.683890	[-0.726657, -0.637061]
Scaled RTN	-0.094253	[-0.104851, -0.084514]
Scaled AW-MSE	-0.067993	[-0.078665, -0.055833]
Affine RTN	-0.491242	[-0.521189, -0.464893]

E MEAN-CENTERING AND REPRESENTATIVE SELECTION

This section compares fixed mean-centering with validation selection and with reconstruction-based selection. Data splits, candidate order and quantizer conventions are defined in Appendix A.

E.1 FIXED CENTERING VERSUS THE FIDELITY-SELECTED SHIFT

Table 16 supplies the selected coefficients and W4 RTN/AW-MSE values for the four original heads. The main-text Table 2 compares these with $t = 1$ for the fragile heads.

Table 16: **Phi benefits most from W4 shifts; the other three heads change little.** RTN and AW-MSE use G128 and independently select coefficients by validation KL. Entries report raw and reparameterized test KL against each head’s source distribution. Gemma 4 ($\dagger$) uses the rank-one soft-cap correction.

Model	t^*	RTN		t^*	AW-MSE	
		Raw KL	Reparam KL		Raw KL	Reparam KL
Phi	4	1.229	0.351	4	0.936	0.256
Gemma 3	0.5	0.049	0.046	−1	0.041	0.038
Gemma 4 $\dagger$	1	0.012	0.010	0.5	0.008	0.008
Qwen3.5	−0.5	0.021	0.019	−0.5	0.013	0.013

Transferring the RTN-selected coefficient to AW-MSE raises Gemma 3 test KL to 0.042, compared with 0.038 under AW-MSE’s own selection and 0.041 without a shift.

E.2 RECONSTRUCTION-SELECTED VERSUS FIDELITY-SELECTED REPRESENTATIVE

The exploratory reconstruction scan uses symmetric RTN at W2, W3 and W4 and $t \in \{0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 2\}$. For $E_t = Q_t - W_t$ and its vocabulary-row mean $\bar{e}_t = E_t^\top \mathbf{1}/V$, it measures squared weight error and its projection onto softmax-visible row differences:

$$R_{\text{raw}}(t) = \|E_t\|_F^2, \quad R_{\text{proj}}(t) = \|PE_t\|_F^2 = R_{\text{raw}}(t) - V\|\bar{e}_t\|_2^2. \quad (10)$$

The implementation divides both quantities by Vd , which leaves their rankings unchanged. This is projected *weight* MSE, not projected logit reconstruction $\mathbb{E}\|PE_th\|^2$ or a Fisher-weighted objective. The removed term measures a shared residual component; it does not supply hidden-state covariance or output-probability weighting.

The retained sweep records cover three models, three precisions and eight coefficients per comparison. Both reconstruction criteria select $t = 1$ in all nine comparisons, while the coefficient minimizing diagnostic KL differs in every case (Table 17). At W4, KL favors $t = 0.5$ for Gemma, $t = 0$ for Qwen and $t = 2$ for Phi within this grid. Projecting away the common residual component therefore does not recover fidelity-based selection.

Table 17: **Weight MSE and KL favor different shifts.** Each row uses the same symmetric RTN quantizer and diagnostic states at every coefficient. Both raw and projected weight MSE select $t = 1$; the KL minimum is descriptive on this diagnostic set, not independently selected on validation.

Model	Bits	Best sampled t			Diagnostic KL	
		MSE	Proj. MSE	KL	At MSE min	At KL min
Gemma 3	4	1	1	0.5	0.0580	0.0465
Gemma 3	3	1	1	0.25	0.3161	0.2340
Gemma 3	2	1	1	0	2.8305	2.5793
Qwen3.5	4	1	1	0	0.0291	0.0202
Qwen3.5	3	1	1	0	0.1599	0.1017
Qwen3.5	2	1	1	0	2.6671	1.8072
Phi	4	1	1	2	0.7981	0.5644
Phi	3	1	1	2	8.9786	5.3672
Phi	2	1	1	1.5	77.0395	74.4024

This is a measured failure of the two weight-space criteria on the tested candidates, not a theorem about all reconstruction objectives. The scan stops at $t = 2$ and cannot locate the later $t = 4$ choice. The wider experiment supplies the separate validation-selected comparison. All rows use 32 test article prefixes and FP32 linear-logit evaluation; per-candidate aggregates are retained, while source and captured-state hashes require additional provenance. Their RTN convention differs from the signed-scale centering control below, so the two tables are not evaluations of identical artifacts.

E.3 RANGE AND SHARED-ROW CONTROLS

A separate signed min-max control compares $t = 0$ and $t = 1$ with BF16 stored scales and reconstructed weights, followed by FP32 logit evaluation. Table 18 reports its W4 results. The shared weight energy is $V\|\mu\|^2/\|W\|_F^2$, and the range ratio compares the mean group maximum absolute weight after and before centering. Smaller ranges alone do not explain the outcome: Gemma and Qwen have smaller mean ranges yet worse KL, whereas Phi has a slightly larger mean range and better KL. Qwen and Phi also have comparable shared weight-energy fractions but opposite fidelity changes.

Table 18: **Smaller group ranges do not guarantee lower KL.** Shared energy and range ratios describe unquantized weights. KL compares raw and mean-centered W4 G128 signed min-max heads on matched diagnostic articles.

Model	Shared weight energy (%)	Range ratio	Raw KL	Centered KL
Gemma 3	3.41	0.9612	0.0499	0.0585
Qwen3.5	14.23	0.9029	0.0196	0.0294
Phi	16.40	1.0128	1.2109	0.8087

F MECHANISM DETAILS

On the matched Phi diagnostics, the selected shift increases reconstruction error while reducing distributional error (Section 4). This appendix defines the probability-bin attribution, tests the residual pattern under stronger GPTQ calibration, and gives a separate cross-model comparison of output sensitivity.

F.1 FISHER AND PROBABILITY-BIN DIAGNOSTICS

The paired diagnostic uses the source-matched Phi bank, its 1,024 fitting states, and the last 16 of its 32 held-out articles (8,176 prediction states). The coefficients $t \in \{0, 1, 4\}$ are fixed; no selection occurs on these states. For each signed W4 G128 RTN/AW-MSE head, the probe retains the quantized-weight hash and computes actual KL and the Fisher quadratic from the same residual $e = E_th$, with FP32 products and TF32 disabled. AW-MSE uses the twelve-factor clipping grid of the probability-probe protocol. The source, hidden-state, fitting-state, artifact and code hashes accompany the measurements.

Both quantizers use signed integers, BF16 stored scales and BF16 reconstructed weights. The RTN convention differs from the symmetric FP32-scale selection probe, so all metrics are recomputed for this comparison.

To understand where the additional reconstruction error goes, we group vocabulary entries by their source probabilities. We then measure how much reconstruction error and Fisher-weighted error each group contributes, using the same groups for every shift.

For each source distribution p , we attribute squared reconstruction energy using $e_c = e - \text{mean}_v(e)\mathbf{1}$ and Fisher energy using $p_v(e_v - p^\top e)^2$. Summing bins recovers $\|Pe\|^2$ and $e^\top F(p)e$, respectively. Error energy in Table 19 is the mean per-state sum within a bin; shares divide aggregate bin energy by aggregate projected energy. Probability mass is averaged across states. The lowest-probability bin contains 98.75% of vocabulary entries on average, so its large error share alone is not evidence of preferential concentration. The evidence is the change in shares across coefficients on the same states, together with the decline in absolute error on high-probability entries. Vocabulary centering removes 4.1% of AW-MSE residual energy at $t = 0$ and 27.5% at $t = 4$; the RTN shares are 0.03% and 5.7%.

Table 19: **Phi’s selected shift reduces error on high-probability outputs.** Bins use unchanged source probabilities on matched states. Error is vocabulary-centered logit-error energy; Fisher shares use probability-centered residuals.

Quantizer	t	Source probability	Mass (%)	Error share (%)	Error energy	Fisher share (%)
RTN	0	$p < 10^{-6}$	0.196	93.474	134039.57	0.317
RTN	0	$10^{-6} \leq p < 10^{-4}$	2.196	6.024	8637.93	4.015
RTN	0	$10^{-4} \leq p < 10^{-2}$	12.842	0.476	682.63	24.223
RTN	0	$p \geq 10^{-2}$	84.766	0.026	36.88	71.444
RTN	1	$p < 10^{-6}$	0.196	94.179	106621.80	0.392
RTN	1	$10^{-6} \leq p < 10^{-4}$	2.196	5.381	6092.47	4.994
RTN	1	$10^{-4} \leq p < 10^{-2}$	12.842	0.421	476.38	28.570
RTN	1	$p \geq 10^{-2}$	84.766	0.019	21.78	66.044
RTN	4	$p < 10^{-6}$	0.196	99.074	395859.36	0.445
RTN	4	$10^{-6} \leq p < 10^{-4}$	2.196	0.862	3444.53	4.941
RTN	4	$10^{-4} \leq p < 10^{-2}$	12.842	0.061	243.17	27.934
RTN	4	$p \geq 10^{-2}$	84.766	0.003	10.37	66.680
AW-MSE	0	$p < 10^{-6}$	0.196	92.436	132569.58	0.348
AW-MSE	0	$10^{-6} \leq p < 10^{-4}$	2.196	7.004	10044.88	4.162
AW-MSE	0	$10^{-4} \leq p < 10^{-2}$	12.842	0.535	767.86	25.173
AW-MSE	0	$p \geq 10^{-2}$	84.766	0.024	35.00	70.316
AW-MSE	1	$p < 10^{-6}$	0.196	92.686	106162.10	0.403
AW-MSE	1	$10^{-6} \leq p < 10^{-4}$	2.196	6.773	7757.36	4.781
AW-MSE	1	$10^{-4} \leq p < 10^{-2}$	12.842	0.518	592.84	27.345
AW-MSE	1	$p \geq 10^{-2}$	84.766	0.023	26.70	67.471
AW-MSE	4	$p < 10^{-6}$	0.196	98.201	288852.93	0.484
AW-MSE	4	$10^{-6} \leq p < 10^{-4}$	2.196	1.679	4938.41	5.228
AW-MSE	4	$10^{-4} \leq p < 10^{-2}$	12.842	0.116	341.06	27.828
AW-MSE	4	$p \geq 10^{-2}$	84.766	0.005	13.48	66.460

F.2 RESIDUAL DIAGNOSTICS UNDER STRONGER GPTQ

We repeat the residual diagnostics for Phi W4 G128 GPTQ using the full Hessian calibrated on 65,536 states, while retaining the original AW-MSE range initializer fitted on 1,024 states. This is the original-range condition in Table 13, with 1% damping and no activation ordering. We compare $t = 0$ with its previously frozen, validation-selected $t^* = 4$; no further coefficient search is performed. Both reconstructed heads exactly match their archived quantized-weight hashes.

The two rows use the same 8,176 held-out states from 16 articles, with FP32 products and log-softmax, TF32 disabled, and BF16 scales and reconstructed weights. These states exactly reproduce the stronger-GPTQ evaluation. The source weights and article identities also match the RTN/AW-MSE probe above, but its older captured hidden-state tensor is not byte-identical. Each comparison is paired within its own capture.

Table 20: **Shifting Phi W4 GPTQ lowers KL despite greater logit-error energy.** GPTQ uses a 65,536-state full Hessian and the original range initializer. Definitions follow Table 4; test states are matched.

t	$\ E_t\ /\ W_t\ $	D_z (%)	Common (%)	Fisher/2	Actual KL
0	0.148515	0.025340	0.4644	0.088771	0.089362
4	0.162178	0.067442	4.7268	0.034710	0.034909

From $t = 0$ to $t = 4$, relative weight error rises by 9.2%, raw logit-error energy rises by $2.66\times$, and vocabulary-centered error energy rises by $2.55\times$. Nevertheless, Fisher-weighted error falls by 60.90% and KL by 60.94%. The quadratic is within 0.7% of actual KL in both rows. Paired article-bootstrap 95% intervals for selected minus raw are $[-0.05975, -0.04954]$ for KL and $[-0.05982, -0.04884]$ for Fisher/2 (10,000 resamples, conditional on the frozen selection). Thus the qualitative residual pattern persists after stronger GPTQ error compensation on this Phi W4 comparison. We do not infer a universal mechanism across models or extend the local quadratic claim to W2.

Table 21: **Phi exceeds Qwen in both logit-error energy and sensitivity.** Unshifted W4 AW-MSE uses disjoint WikiText articles (16,352 positions/model). D_z normalizes error by source energy; S_p measures sensitivity per unit error and C_{out} its Fisher approximation on the same residuals.

Model	$\mathbb{E}\ \delta z\ ^2$	D_z	S_p	C_{out}	KL
Gemma 3 4B	2.08×10^4	2.93×10^{-3}	3.89×10^{-6}	3.94×10^{-6}	0.041
Qwen3.5 4B	1.23×10^4	7.44×10^{-3}	2.09×10^{-6}	2.09×10^{-6}	0.013
Phi-4-mini	1.47×10^5	2.41×10^{-3}	1.26×10^{-5}	1.19×10^{-5}	0.925

F.3 CROSS-MODEL OUTPUT SENSITIVITY

Weight-space error is not a reliable proxy for prediction fidelity. On a matched Gemma W3 G128 control, activation weighting *raises* relative weight L_2 error from 0.2063 to 0.2162 while *reducing* held-out KL from 0.24443 to 0.14194: weight-space distortion becomes logit error only through the occupied hidden-state directions, and the prediction distribution reweights that error. This motivates selecting representatives by prediction fidelity rather than by weight norm.

For the cross-model comparison, let $\delta z = (Q - W)h$ and use the categorical Fisher matrix and local KL expansion from Equation 6. For nonzero aggregate error, the per-unit sensitivity is $C_{\text{out}} = \mathbb{E}[\delta z^\top F(p)\delta z] / \mathbb{E}\|\delta z\|_2^2$, and $S_p = 2\mathbb{E}[D_{\text{KL}}] / \mathbb{E}\|\delta z\|_2^2$ is the actual KL per unit logit error. Table 21 compares W4 AW-MSE heads fitted to the same articles across Gemma, Qwen and Phi, with source-normalized distortion $D_z = \mathbb{E}\|\delta z\|^2 / \mathbb{E}\|z\|^2$. Phi has smaller source-normalized distortion D_z but roughly $12\times$ larger absolute logit-error energy and about $6\times$ higher per-unit Fisher sensitivity than Qwen; both factors contribute to its much larger KL. Phi also has a smaller mean top-two logit margin than Gemma (1.96 versus 2.59) and a larger mean Fisher trace (0.592 versus 0.452). These measurements describe the evaluated unshifted heads; they do not establish a general predictor of the benefit from reparameterization.

G DEPLOYMENT AND SYSTEMS VALIDATION

This section gives the matched packed Phi export and timing protocol behind the main deployment table, the tied/untied storage accounting, and an auxiliary Qwen serving measurement. Packed checkpoints use the original parameterization unless the reparameterized export is stated; each numerical path retains its own BF16 reference.

G.1 MATCHED PHI PACKED EXPORT AND TIMING

The deployment retest starts from one BF16 Phi source and independently builds untied BF16 heads at $t = 0, 4$ and W4 G128 heads at $t = 0, 4$ for min-max and AW-MSE. The exporter resolves the source output matrix, preserves all 194 non-head tensors byte-for-byte, and verifies every serialized head tensor. For the tied source, the BF16 input embedding is retained. Every W4 log records selection of the Marlin linear kernel in vLLM 0.28.0 on an A10G.

Quality uses 32 contiguous 512-token WikiText blocks, scoring 16,352 next tokens with eager execution and a 1,024-token model limit. Corpus and token hashes match across all seven rows. Per-block sums reproduce NLL and its exponential reproduces PPL. These are the historical deployment blocks, not the article-partitioned selection/evaluation split and not a fresh holdout. Table 22 isolates the BF16 serialization controls.

Table 22: **Untying and BF16 serialization leave small likelihood differences on Phi.** Matched exports test the unshifted and shifted heads; algebraic softmax invariance does not imply bitwise inference equivalence.

BF16 head	WikiText PPL	$\Delta\text{NLL vs. source}$
Tied source	11.6477	+0.000000
Untied, $t = 0$	11.6455	-0.000194
Untied, $t = 4$	11.6363	-0.000983

The selected $t = 4$ is fixed before deployment evaluation for both base quantizers. AW-MSE uses second moments from 1,024 archived fitting states whose source checkpoint identity matches the deployment reference. The exporter computes the row mean and shift in FP32, casts the representative to BF16, then selects signed-integer codes using floating-point candidate scales and stores BF16 scales. Its clipping grid is $\{1, .975, .95, .925, .9, .875, .85, .8, .75, .7\}$. This is a documented deployment-transfer path; it differs from the probability probes’ scale-rounding and fitting conventions. No coefficient or clipping grid is selected using the reported deployment PPL.

Timing uses greedy 64-token generation at batches one and 16, with prefix caching disabled and normal graph execution. Each cell has three warmups and seven timed repetitions. We independently reload the five source/W4 checkpoints in two rounds, reversing their order in the second round. The main table reports the mean of the two round medians; timing includes prefill and generation and excludes loading. GPU access is exclusive during the run. Checkpoint hashes match between every quality and timing row. Shifted AW-MSE batch-one medians are 1,013.24 and 1,013.53 ms; its B16 medians are 1,210.19 and 1,210.39 ms. These repeated measurements show essentially unchanged W4 latency, not a statistical guarantee of zero overhead on other workloads or devices. The matched min–max exports reproduce the 40.57 to 16.28 PPL reduction. A deployed rank-one correction for Gemma 4 is not part of these measurements.

GPTQ construction and timing scope. The deployed full-Hessian GPTQ head uses the same INT4 Marlin format as AW-MSE. Its construction takes about one minute per shifted head, compared with a few seconds for AW-MSE scale selection; Appendix A.1 reports full sweep costs. GPTQ PPL and KL are evaluated separately. GPTQ was not timed, so its latency entries in Table 3 are left unreported.

G.2 TIED AND UNTIED STORAGE

Head payload. With Vd indices and one BF16 scale per G weights, the dominant payload is

$$M_{\text{head}}(b, G) = Vd \left(\frac{b}{8} + \frac{2}{G} \right) \text{ bytes.} \quad (11)$$

This excludes small metadata and does not account for any retained embedding.

Model interface. A model adapter identifies the output matrix, final normalized hidden states, embedding-sharing configuration, and any transformation between linear logits and probabilities. For an already untied head, the packed output matrix replaces the existing head. For a tied head, the evaluated layout retains the original BF16 embedding and adds an independently packed output projection. A runtime that supports a shared quantized embedding and projection would require separate evaluation of both operators and their combined quality impact. The current head-only comparisons do not evaluate that alternative.

Storage and memory traffic. Table 23 instantiates Equation 11 for the evaluated checkpoint. In an untied model, replacing an independent BF16 head reduces its weight storage by $2Vd - M_{\text{head}}$. In the evaluated tied layout, preserving the BF16 embedding instead adds M_{head} of resident weights. The tied Phi source loads about 7.17 GiB of model weights, versus 7.47 GiB with the independent packed head: lower output-projection traffic coexists with higher resident weight memory. Actual memory traffic depends on caching, batch reuse, packing and kernel execution; latency also includes the decoder, output processing and sampling. The payload calculation does not measure runtime workspaces or the KV cache.

Packing and loading. For the packed W4/W8 configurations, the checkpoint builder stores signed quantized codes in INT32 containers with $32/b$ codes per container, together with BF16 group scales and the original matrix shape. It exports an independent compressed-tensors head group and uses the target pattern `re:.*lm_head$` to match the mapped output operator. Tied-model export clears the applicable embedding-sharing flags while retaining the source input table. A deployment audit should bind the fitted artifact, serialized tensors and loaded operator to the same weights and stored scales, verify the signed code convention through a packing round trip, and record actual kernel dispatch. The intended serving path uses vLLM and a compatible packed projection such as Marlin (Kwon

Table 23: **W4 G128 uses 26% of the BF16 head payload.** Analytical Gemma 3 payload: $V = 262,208$, $d = 2,560$. An untied packed head adds resident storage when the BF16 input embedding is retained.

Representation	Head payload (GiB)	Relative to BF16
BF16	1.2503	1.0000
W8, G128	0.6349	0.5078
W4, G128	0.3223	0.2578
W4, G32	0.3516	0.2812

et al., 2023; IST-DASLab, 2025); quantization metadata or reduced file size alone does not establish that this path ran.

G.3 AUXILIARY QWEN SERVING

The Qwen3.5 4B and Phi-4-mini checkpoints used above quantize BF16-decoder heads, retain BF16 input embeddings and export independent W4/W8 G128 heads. Fitting uses 128 generic prompts with eight sampled positions each. The Qwen and Phi KL and top-1 diagnostics use the 1,024 fitting states: at W4, AW-MSE improves Qwen top-1 agreement from 93.26% to 95.21% and Phi from 37.01% to 49.12% (in-distribution diagnostics on the fitting states, without exhaustive non-head tensor audits).

Table 24: **W4 reduces Qwen3.5 4B batch-one latency by 9.6% versus BF16 on A10G.** Head-only W4/W8 use G128; timings cover 25-token greedy generation over seven warmed compiled-vLLM repetitions, excluding loading and compilation.

Head	B1 latency (ms)	B16 throughput (tokens/s)
BF16	516.6	570.9
W8 AW-MSE	480.4	585.9
W4 AW-MSE	467.2	614.8

Table 24 reports medians from persistent engines. The kernel audit records vLLM 0.28.0 with compiled CUDA graphs and MarlinLinearKernel for Qwen’s packed heads. Generation timing includes the output projection and surrounding model execution, rather than an isolated head microbenchmark; the 9.6% batch-one latency reduction is $1 - 467.2/516.6$ using unrounded measurements. Warm persistent engines exclude loading and compilation, and the interval does not separately identify prefill and decode costs or variation across independent engine starts.

H MODEL, HEAD AND VOCABULARY DETAILS

These tables support Figure 1 and the cross-model precision comparison; they are reference material rather than part of the method.

H.1 OUTPUT-HEAD SIZES

Table 25 gives the dimensions of the evaluated heads. Head parameters are Vd ; the nominal share divides this count by the model size in the name (3.8B for Phi-4-mini). It is a scale indicator, not a fraction of resident memory. Tied embeddings and multimodal components require separate accounting. Gemma 4 E4B is included for its output dimensions, without assigning a nominal share: its effective-size designation is not a total parameter count. The Gemma 3 row uses the evaluated checkpoint’s vocabulary dimension; the other rows use official configurations (Google, 2026; Qwen Team, 2026; Microsoft, 2025). The three additional very-large-vocabulary heads (BLOOM, BLOOMZ, XGLM) are 1.7B models whose 250–256K vocabularies place roughly 30% of nominal parameters in the output head.

Vocabulary comparison and sources. Figure 1 compares 23 selected model families using output-matrix row counts, including reserved or padded IDs. Earlier bars are shown for 16 families; seven

Table 25: Output-projection dimensions and nominal parameter shares. Gemma 4’s 262,144-row configuration and the evaluated Gemma 3 checkpoint’s 262,208-row matrix both round to 671M head weights.

Model	Vocabulary	Width	Head (M)	Nominal share (%)
Gemma 3 4B	262,208	2,560	671	16.8
Gemma 4 E4B	262,144	2,560	671	—
Qwen3.5 4B	248,320	2,560	636	15.9
Phi-4-mini	200,064	3,072	615	16.2
BLOOM-1.7B	250,880	2,048	514	30.2
BLOOMZ-1.7B	250,880	2,048	514	30.2
XGLM-1.7B	256,008	2,048	524	30.8

have only a selected checkpoint. The paired examples do not establish monotonic growth across intermediate releases. The GPT row groups a vendor lineage across architectural changes. The right panel uses each selected checkpoint’s own hidden width and decimal GB; its fourfold reduction describes raw weight codes only. Table 26 identifies all 23 selected checkpoints and the 16 earlier references, with linked configurations and release sources. The Gemma 3 row uses the evaluated 262,208-row matrix from Table 25; native Gemma configurations can use 262,144. Machine-readable plotted values and the full source record accompany the paper. Table 27 provides additional intermediate generations for four families. Each is read from the model’s official configuration (the `vocab_size` field of `config.json` on the Hugging Face Hub); the corresponding family reports and cards include (Grattafiori et al., 2024; Qwen Team, Alibaba Cloud, 2024; Qwen Team, 2026; Microsoft, 2025; Mistral AI, 2025).

Table 26: Checkpoints behind Figure 1, in plotted order. Checkpoint names link to configuration sources and years to release sources. *V* counts output rows; *d* is the selected checkpoint width. A dash indicates no earlier comparison.

Family	Earlier checkpoint and rows	Selected checkpoint and rows	<i>d</i>
Gemma	google/gemma-2b [2024]; <i>V</i> = 256,000	google/gemma-3-4b-it [2025]; <i>V</i> = 262,208	2560
XGLM	—	facebook/xglm-1.7B [2021]; <i>V</i> = 256,008	2048
BLOOM	—	bigscience/bloom-1b7 [2022]; <i>V</i> = 250,880	2048
Qwen	Qwen/Qwen-7B [2023]; <i>V</i> = 151,936	Qwen/Qwen3.5-4B [2026]; <i>V</i> = 248,320	2560
Llama	meta-llama/Llama-2-7b-hf [2023]; <i>V</i> = 32,000	meta-llama/Llama-4-Scout-17B-16E-Instruct [2025]; <i>V</i> = 202,048	5120
GPT	openai-community/gpt2 [2019]; <i>V</i> = 50,257	openai/gpt-oss-20b [2025]; <i>V</i> = 201,088	2880
Phi	microsoft/phi-2 [2023]; <i>V</i> = 51,200	microsoft/Phi-4-mini-instruct [2025]; <i>V</i> = 200,064	3072
GLM	THUDM/chatglm-6b [2023]; <i>V</i> = 130,528	zai-org/GLM-4-9B-0414 [2025]; <i>V</i> = 151,552	4096
Falcon	tiiuae/falcon-7b [2023]; <i>V</i> = 65,024	tiiuae/Falcon3-7B-Base [2024]; <i>V</i> = 131,072	3072
Mistral	mistralai/Mistral-7B-v0.1 [2023]; <i>V</i> = 32,000	mistralai/Mistral-3-3B-Instruct-2512-BF16 [2025]; <i>V</i> = 131,072	3072
DeepSeek	deepseek-ai/deepseek-llm-7b-base [2023]; <i>V</i> = 102,400	deepseek-ai/DeepSeek-V3 [2024]; <i>V</i> = 129,280	7168
InternLM	internlm/internlm-7b [2023]; <i>V</i> = 103,168	internlm/internlm3-8b-instruct [2025]; <i>V</i> = 128,512	4096
SmolLM	HuggingFaceTB/SmolLM-1.7B [2024]; <i>V</i> = 49,152	HuggingFaceTB/SmolLM3-3B [2025]; <i>V</i> = 128,256	2048
Hunyuan	—	tencent/Hunyuan-A13B-Instruct [2025]; <i>V</i> = 128,167	4096
Baichuan	baichuan-inc/Baichuan-7B [2023]; <i>V</i> = 64,000	baichuan-inc/Baichuan2-7B-Base [2023]; <i>V</i> = 125,696	4096
Granite	ibm-granite/granite-3.0-2b-base [2024]; <i>V</i> = 49,152	ibm-granite/granite-4.0-micro [2025]; <i>V</i> = 100,352	2560

Continued on next page

Table 26 (continued)

Family	Earlier checkpoint and rows	Selected checkpoint and rows	d
StableLM	stabilityai/stablelm-base-alpha-3b [2023]; $V = 50,688$	stabilityai/stablelm-2-1_6b [2024]; $V = 100,352$	2048
OLMo	allenai/OLMo-7B-hf [2024]; $V = 50,304$	allenai/Olmo-3-7B-Think [2025]; $V = 100,278$	4096
RWKV	RWKV/rwkv-4-169m-pile [2022]; $V = 50,277$	RWKV/rwkv-5-world-1b5 [2023]; $V = 65,536$	2048
GPT-J	—	EleutherAI/gpt-j-6b [2021]; $V = 50,400$	4096
OPT	—	facebook/opt-6.7b [2022]; $V = 50,272$	4096
StarCoder	—	bigcode/starcoder2-3b [2024]; $V = 49,152$	3072
OpenELM	—	apple/OpenELM-3B [2024]; $V = 32,000$	3072

Table 27: Vocabulary sizes across selected generations of four families, from each model’s official configuration.

Family	Model	Vocabulary
Phi	Phi-2	51,200
	Phi-3-mini	32,064
	Phi-4	100,352
	Phi-4-mini	200,064
Llama	Llama 2	32,000
	Llama 3	128,256
	Llama 4	202,048
Qwen	Qwen1-2.5	151,936
	Qwen3.5	248,320
Mistral	Mistral 7B	32,000
	Ministral 3	131,072

H.2 PRECISION AND PERPLEXITY ACROSS MODELS

Similarly sized heads have different precision requirements (Table 28). W4 AW-MSE keeps perplexity within 1.1% of each model’s BF16 reference for Gemma 3/4 and Qwen3.5. Phi is the exception: weighting reduces its W4 PPL from 40.57 to 30.94, still far above BF16’s 11.65; W8 AW-MSE reaches 11.75. Among the evaluated settings, W8 AW-MSE preserves near-reference likelihood for Phi, while W4 AW-MSE suffices for the other three models. The Qwen-Phi contrast holds group size fixed and establishes that matrix size alone does not determine the observed precision requirement; distributional diagnostics on their fitting states show the same contrast (W4 AW-MSE KL 0.01361 for Qwen versus 0.88979 for Phi), examined further in Appendix F.3. These observations compare practical operating points rather than locating each model’s lowest viable precision.

I CONFIDENCE AND LIKELIHOOD CONTROLS

These unshifted-head controls distinguish preservation of source predictions from incidental changes in confidence; they do not select a reparameterization coefficient. Raw perplexity can reward a change in confidence rather than improved source fidelity: in the original Gemma precision sweep, fitting separate source and candidate temperatures on a held-out block half reverses apparent likelihood gains (W4 AW-MSE excess NLL $-0.00364 \rightarrow +0.04745$ nats/token; W3 $-0.04193 \rightarrow +0.10236$), consistent with quantization softening an overconfident source while adding distortion (Guo et al., 2017). This is why we select representatives by source-to-candidate KL and report perplexity separately. These controls use the unshifted precision-sweep artifacts; matched temperature controls for the reparameterized shifted heads remain to be measured and are not substituted by the original-head controls.

Table 28: **W4 AW-MSE retains near-BF16 perplexity for three heads but substantially degrades Phi.** Head-only WikiText evaluation uses G32 for Gemma 3 W4 and G128 otherwise. Each model uses its own BF16 reference and tokenizer, with matched Hugging Face loading and evaluation paths (Appendix G.3).

Model	Head (M)	BF16	W4 min-max	W4 AW-MSE	W8 AW-MSE
Gemma 3 4B	671	60.12	60.84	59.46	60.12
Gemma 4 E4B	671	74.97	75.48	75.55	74.95
Qwen3.5 4B	636	10.89	11.19	10.98	10.89
Phi-4-mini	615	11.65	40.58	30.94	11.75

J HIGHER-CAPACITY PARAMETERIZATION OF THE EQUIVALENCE CLASS

J.1 PARAMETERIZATION AND SEARCH

The scalar parameterization can be generalized by assigning a separate coefficient to each quantization group. Let $g(j)$ denote the group containing hidden dimension j . We define

$$W_{v,j}^{\text{grp}} = W_{v,j} - t_{g(j)}\mu_j. \quad (12)$$

Equivalently, $W^{\text{grp}} = W - \mathbf{1}a^\top$ with $a_j = t_{g(j)}\mu_j$, so the full-precision softmax distribution remains exactly unchanged for a linear-softmax head. Aligning the coefficients with quantization groups allows independently quantized regions of the head to select different shift magnitudes while retaining the same packed layout and, for shift-compatible logit paths, no additional inference operation.

The groups partition the hidden dimension; every vocabulary row receives the same vector subtraction. With contiguous 128-channel groups, SmolLM3’s 2,048-wide head has 16 coefficients. Setting every coefficient to the same t recovers the scalar family. The extension enlarges the search space while the matrix correction $\mathbf{1}a^\top$ remains rank one. Nonlinear paths use the correction in Appendix B, restoring $(a^\top h)\mathbf{1}$ before the logit transformation.

Optimization. We initialize all group coefficients at the validation-selected shared t^* and perform coordinate-wise search over groups using validation KL. At each update, we vary one coefficient while holding the others fixed and retain a proposal only if it lowers full-vocabulary validation KL. The coefficients are selected jointly through this objective, separately for each base quantizer; they are frozen before test evaluation.

Groups are visited in descending order of $\sum_{j \in g} m_j \mu_j^2$, using fitting moments $m_j = \mathbb{E}_{\text{fit}}[h_j^2]$ only. We perform 32 coordinate updates, each evaluating positive and negative proposals from the current coefficient vector. The lower-KL proposal is accepted only if it improves on the current vector. Step sizes are 0.5, 0.125 and 0.03125 on successive passes through the ordered groups, retaining the last step size for later passes. The evaluated heads have 16–24 groups, so 32 updates use only the first two step sizes and may stop partway through the second pass. Coefficients are not clipped. GPTQ performs its own search, using its fitting Hessian and validation objective.

We report two experiments with different search budgets and evaluation sets (Table 29). Both initialization grids include zero shift. The nine-model experiment extends the paper’s 14-point scalar search with 64 grouped proposals. The separate SmolLM3 pilot compares grouped search with scalar refinement using the same total candidate count. These finite searches do not establish an optimal shift.

Table 29: Grouped-search protocols. Counts are validation evaluations per quantizer. Each grouped search adds 32 updates with two proposals each.

	Nine-model experiment	SmolLM3 pilot
Initialization	Original 14-point scalar grid	Shared 27-point scalar grid
Scalar budget	14	27 + 64 = 91 (scalar refinement)
Grouped budget	14 + 64 = 78	27 + 64 = 91
Evaluation	Original 16 test articles	16 previously unused articles
Quantizers	RTN, AW-MSE, GPTQ	RTN, AW-MSE

J.2 EVALUATION ACROSS NINE MODELS

We evaluate W4 G128 heads from Phi-4-mini, Gemma 3/4, Qwen3.5, Qwen3, Ministral, BLOOM, BLOOMZ and XGLM. The BF16 decoder remains fixed. English WikiText supplies 128 fitting articles with eight states each, 16 selection articles and 16 disjoint test articles, using 512-token prefixes. All choices are frozen before test evaluation. This exploratory follow-up reuses the original paper test articles; it is not an untouched replication.

Table 30 reports matched raw, scalar and grouped results rerun together on H100. Grouped search lowers test KL on all nine models under RTN and AW-MSE, with reductions of 3.0–20.3% and 2.3–22.2%, respectively. For each quantizer, eight of nine paired intervals exclude zero. GPTQ gains are smaller and mixed: five models improve, four regress slightly, and only Phi’s interval excludes zero. The additional validation budget prevents attributing these gains solely to the larger parameterization.

Table 30: Grouped search lowers RTN/AW-MSE KL on all nine models; GPTQ results are mixed. W4 G128 KL uses each model’s source; scalar/grouped budgets are 14/78 validation evaluations. Positive reductions favor grouped search; $\Delta = \text{KL}_{\text{grp}} - \text{KL}_{\text{scalar}}$. Pointwise paired article-bootstrap 95% intervals use 10,000 resamples, conditional on frozen selections, on previously examined test articles. Gemma 4 ($\dagger$) restores the common offset before soft-capping.

Model	Quantizer	t^*	Raw	Scalar	Grouped	Red. (%)	Δ KL: 95% CI
Phi-4-mini	RTN	4	1.230021	0.350579	0.300249	+14.36	[−0.059382, −0.042322]
	AW-MSE	4	0.939398	0.255508	0.222474	+12.93	[−0.040437, −0.026940]
	GPTQ	4	0.160414	0.060511	0.058634	+3.10	[−0.003556, −0.000413]
Gemma 3	RTN	0.5	0.049208	0.046005	0.038335	+16.67	[−0.009860, −0.005429]
	AW-MSE	−1	0.040232	0.037523	0.033956	+9.51	[−0.005063, −0.002085]
	GPTQ	0	0.033609	0.033609	0.032296	+3.91	[−0.003080, +0.000345]
Gemma 4 †	RTN	1	0.011617	0.010587	0.010273	+2.97	[−0.000662, +0.000044]
	AW-MSE	0.5	0.008402	0.008248	0.007313	+11.33	[−0.001141, −0.000728]
	GPTQ	1	0.008423	0.008132	0.008329	−2.42	[−0.000021, +0.000399]
Qwen3.5	RTN	−0.5	0.020831	0.018588	0.016096	+13.41	[−0.002920, −0.002062]
	AW-MSE	−0.5	0.013097	0.013075	0.011662	+10.81	[−0.001813, −0.001038]
	GPTQ	−0.5	0.010844	0.010470	0.010629	−1.52	[−0.000146, +0.000456]
Qwen3	RTN	−0.5	0.028519	0.027787	0.023311	+16.11	[−0.005166, −0.003727]
	AW-MSE	0	0.013296	0.013296	0.011808	+11.19	[−0.001841, −0.001141]
	GPTQ	0	0.007929	0.007929	0.008129	−2.52	[−0.000079, +0.000477]
Ministral	RTN	0	0.007443	0.007443	0.007061	+5.13	[−0.000534, −0.000251]
	AW-MSE	0	0.005576	0.005576	0.005450	+2.26	[−0.000284, +0.000053]
	GPTQ	0	0.006135	0.006135	0.006097	+0.63	[−0.000178, +0.000093]
BLOOM	RTN	6	1.116508	0.531289	0.423429	+20.30	[−0.136209, −0.085733]
	AW-MSE	4	0.594330	0.136469	0.121634	+10.87	[−0.018924, −0.011113]
	GPTQ	4	0.036659	0.028343	0.028132	+0.74	[−0.001123, +0.000791]
BLOOMZ	RTN	4	1.119557	0.483042	0.385792	+20.13	[−0.109115, −0.085646]
	AW-MSE	4	0.662490	0.157715	0.122734	+22.18	[−0.039459, −0.029717]
	GPTQ	4	0.041895	0.032715	0.032838	−0.38	[−0.000105, +0.000337]
XGLM	RTN	1	2.128650	0.143396	0.127086	+11.37	[−0.022235, −0.011954]
	AW-MSE	1	0.586145	0.094916	0.085099	+10.34	[−0.013016, −0.006262]
	GPTQ	1	0.008968	0.006694	0.006691	+0.04	[−0.000136, +0.000141]

All metrics use reconstructed heads, full-vocabulary FP32 projection and log-softmax on frozen BF16 states, with TF32 disabled. RTN uses signed $-7, \dots, 7$ codes and FP32 scales; AW-MSE and GPTQ use signed $-8, \dots, 7$ codes, the 12-point clipping grid, and BF16 scales and reconstruction. GPTQ uses 1% damping and no activation ordering (Appendix A). Gemma 4 uses the original pretrained checkpoint and restores $(a^\top h)1$ before its soft-cap with threshold 30. These results do not measure packed-serving latency.

Table 31 gives perplexity on the same test states. KL reduction does not imply lower perplexity in every configuration; for example, Gemma 3 AW-MSE improves KL while slightly increasing perplexity. The source references and numerical path belong to this comparison and should not be pooled with the packed-deployment measurements.

Table 31: **Lower KL does not always yield lower perplexity.** Nine-model scalar $\rightarrow$ grouped perplexity uses the same states as Table 30. Source PPL is each model’s BF16 reference.

Model	Source PPL	Scalar $\rightarrow$ grouped PPL		
		RTN	AW-MSE	GPTQ
Phi-4-mini	9.719	13.461 $\rightarrow$ 13.042	12.486 $\rightarrow$ 12.106	10.286 $\rightarrow$ 10.287
Gemma 3	37.977	38.482 $\rightarrow$ 38.419	38.086 $\rightarrow$ 38.176	38.852 $\rightarrow$ 38.613
Gemma 4 [†]	67.223	68.156 $\rightarrow$ 68.011	68.192 $\rightarrow$ 68.129	67.646 $\rightarrow$ 67.961
Qwen3.5	8.977	9.131 $\rightarrow$ 9.143	9.106 $\rightarrow$ 9.060	9.065 $\rightarrow$ 9.054
Qwen3	12.577	13.020 $\rightarrow$ 12.839	12.747 $\rightarrow$ 12.617	12.636 $\rightarrow$ 12.639
Ministral	8.598	8.629 $\rightarrow$ 8.633	8.665 $\rightarrow$ 8.662	8.660 $\rightarrow$ 8.632
BLOOM	18.469	29.826 $\rightarrow$ 27.845	21.030 $\rightarrow$ 20.774	18.868 $\rightarrow$ 18.864
BLOOMZ	22.067	31.237 $\rightarrow$ 29.965	25.271 $\rightarrow$ 25.145	22.745 $\rightarrow$ 22.670
XGLM	13.809	15.877 $\rightarrow$ 15.712	15.483 $\rightarrow$ 15.120	13.886 $\rightarrow$ 13.862

J.3 SMOLLM3 COMPARISON WITH MATCHED CANDIDATE COUNTS

The separate SmoLLM3 pilot initializes both searches from 27 scalar candidates. Grouped search spends 64 additional evaluations on the coordinate proposals above; the scalar baseline spends 64 on finer one-dimensional refinement. Both therefore use 91 validation evaluations, although their search regions and optimization procedures differ.

On 16 previously unused English WikiText articles, the frozen grouped shift reduces SmoLLM3 W4 G128 KL from 0.8923 to 0.8008 under RTN and from 0.3404 to 0.2848 under AW-MSE, relative to the refined scalar search. These articles come from the WikiText validation corpus split and exclude the original fitting, selection and test articles. Paired article-bootstrap 95% intervals for $KL_{\text{grp}} - KL_{\text{LD}}$ are $[-0.1232, -0.0630]$ and $[-0.0702, -0.0414]$, respectively (10,000 resamples). These are pointwise intervals for one model in an exploratory follow-up. The measurements use reconstructed heads on matched frozen decoder states; packed-serving latency was not measured for the grouped extension.